# Inaccessibility and Compliance: Geography, Institutions, and Agency in Foreign Aid*

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**Abstract**

Existing scholarship highlights that bureaucrats from aid agencies and international organizations are crucial for addressing international development challenges, but international bureaucrats' impacts are often conditional on structural constraints. We argue that when geographic and institutional constraints align, bureaucratic behavior follows a rational inattention framework. In these contexts, career-minded international bureaucrats, who are cost- and information-constrained, allocate finite attention to seemingly more solvable problems. We test our pre-registered hypotheses using data on project-level compliance with World Bank safeguard policies on involuntary resettlement, indigenous peoples, and the environment. Using spatial regressions that account for autocorrelation, we find that longer travel times to monitor projects negatively affect safeguard compliance. Consistent with rational inattention, effects concentrate among the most inaccessible projects. Project density that does not account for road networks—or states' infrastructural power—is also less predictive of overall patterns. While interaction analyses show that higher-performing international bureaucrats can partially mitigate the structural constraints of geography and institutions, structures dominate. Our results demonstrate how the interplay of geography and institutions channels bureaucratic behavior, clarifying the limits of individual agency in development.

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Does geography structure international development outcomes? Prominent answers from Acemoglu and Robinson (2012), Collier (2007), Diamond (1999), Rodrik, Subramanian and Trebbi (2004) and Sachs (2005) debate the relative roles of geography and institutions. Largely, these debates focus on differences in cross-national factors, such as disease environments, natural resources, and slow-moving institutions (e.g., property rights and rule of law). Yet *within countries*, geography and institutions often intertwine to reinforce structural constraints (Acemoglu, García-Jimeno and Robinson, 2015; Gerring and Christenson, 2017, 65). In particular, the combination of geography and institutions explains subnational variation in state reach—or what Mann (1984) famously terms "infrastructural power".

In this paper, we examine the effects of within-country alignment between geographic and institutional constraints on foreign aid implementation. The special issue framework paper suggests that such settings generate "conforming alignment" between structural constraints and agents' behavior (Weaver, Morrison and Heinzel, 2026). We extend these insights by drawing from work on rational inattention in macroeconomic policy (Sims, 2003). Given international bureaucrats' career ambitions, as well as their cost and information constraints, we argue that conforming alignment spurs bureaucrats to allocate finite attention toward problems that seem more solvable and legible to principals. However, higher-performing bureaucrats from foreign aid agencies and international organizations can separate signal from noise and at least partly address more severe recipient-country structural constraints.

We examine the relevance of our argument in project-level compliance with the World Bank's social and environmental safeguard policies on involuntary resettlement, indigenous peoples, and the environment. When there is noncompliance with these risk management policies, the resulting negative externalities can involve the loss of life and severe livelihood destruction (Buntaine, 2016). That is why all major aid donors nowadays have social and environmental safeguard policies—even if compliance with them is uneven (Greenstein, 2022).

To measure the constraints posed by the combination of geographic and institutional

structures, we use data on (log) driving times from OpenStreetMap to capture which aid projects are most difficult to access and monitor. Our measure combines the driving times from (1) the airport to each project's most accessible location; and (2) that project's location to the nearest location of another project, which differs on a yearly basis as new projects start and close. We incorporate the airport time to mimic a bureaucratic supervision mission from abroad, and we use nearest-neighbor project travel time to capture how structural pressures change over time. As an alternative measure of structures that is more related to geography than institutions, we perform spatial grid-cell analysis to map the density of nearby projects, irrespective of the road network. Then, we consider how results change when we add first- and second-order neighboring areas as well.[1]

To complement the above analyses that focus on the structural constraints to development, we undertake more agency-centric analyses as well. We do so by interacting the log driving time variable with Denly's (2025) measure of World Bank Task Team Leaders (TTL) performance. Given that TTLs are the primary bureaucrats in charge of supervising each project, the interaction captures whether higher-performing international bureaucrats can overcome more domestically-related, structural development challenges.

Consistent with our pre-analysis plan,[2] we find strong statistical support for the hypothesis that larger (log) driving times yield lower levels of safeguard policy compliance. In line with our theoretical expectation concerning rational inattention, effects concentrate in the upper tail of the distribution, corresponding to the most inaccessible projects. Analyses of nearby projects indicate that higher project density increases compliance directionally but not consistently. These results suggest that geographic density itself is insufficient without roads and state presence. The interaction analyses also indicate that higher-performing TTLs can partially mitigate structural development challenges. However, even at the highest observed levels of TTL performance, inaccessibility constrains compliance efforts that aim

[1] Technically, first- and second-order neighboring areas refer to first- and second-order spatial lags, which we describe later in the paper.

[2] See: https://osf.io/ru9dp/?view_only=e7458ea391bd4a90b7ffae2b5baba451

to protect individuals and the environment from the negative externalities of development.

These findings help clarify the limits of international bureaucrats' agency. In line with principal-agent and public choice accounts, even mission-driven bureaucrats face career and monitoring incentives that make it risky to invest time into seemingly hard-to-solve projects. That is especially the case where principals at aid agencies and international organizations can more easily punish visible failures than reward invisible prevention. The broader implication is that mission motivation and professional norms shape what bureaucrats aim to achieve, and in many cases they are successful (Honig, 2018; Heinzel and Liese, 2021; Limodio, 2021). Nevertheless, incentives, information, and monitoring constraints also shape what bureaucrats can reliably accomplish (Khan, Khwaja and Olken, 2016, 2019; Dhaliwal and Hanna, 2017; Williams, 2017; Bertrand et al., 2020; Dasgupta and Kapur, 2020; Bandiera et al., 2021). In making these distinctions with our rational inattention lens, we aim to reconcile more incentive-based and sociological accounts about bureaucrats.

The results also inform debates about the role of geography in international development. Our results suggest that favorable geography is insufficient on its own to consistently mitigate issues in foreign aid implementation. In line with Acemoglu, García-Jimeno and Robinson (2015), it is the combination of geography and institutions, taking the form of state infrastructural power, that reduces negative social and environmental externalities. These patterns persist even after accounting for local-level population measures, underscoring the interplay of geography and institutions, not contested accounts of population density (e.g., Herbst, 2000; Acemoglu, Johnson and Robinson, 2002; Bandyopadhyay and Green, 2012). Our results thus align with other studies emphasizing complex dependencies between geography and institutions (Sokoloff and Engerman, 2000; Easterly and Levine, 2003; Nunn and Puga, 2012; Alsan, 2015; Mayshar, Moav and Neeman, 2017).

Finally, our results add nuance to the literature on whether "good countries" or "good projects" are more important for fostering aid effectiveness outcomes. Denizer, Kaufmann and Kraay (2013) and Bulman, Kolkma and Kraay (2017) provide the headline findings.

They highlight that 75–90% of the variation in project-level aid effectiveness outcomes reflect within-country features.[3] These headline findings are consistent with the present paper's focus on within-country phenomena, but the broader take-away is different. In the debate on "good countries" versus "good projects", the latter generally refer to country-specific design and implementation choices for which aid providers have agency. However, aid providers usually engage in complicated negotiations with recipients (Swedlund, 2017; Bennon and Fukuyama, 2022), and providers generally do not have full control over which locations within a country receive aid (Hodler and Raschky, 2014; Briggs, 2017; Dreher et al., 2019; Song, Brazys and Vadlamannati, 2021). Consequently, within-country structural problems still underpin much of the foreign aid literature that tackles bureaucratic autonomy and agency (e.g., Honig, 2018). Our paper suggests a further role for how international bureaucrats address structure-agency trade-offs in explaining compliance outcomes.

# 1. Theory

In this paper, we develop a theory of compliance that integrates macro-structural constraints and micro-founded bureaucratic behaviors. Rather than adjudicating between the canonical theories of compliance (see Table 1), we use them as complementary and helpful inputs to construct an account focusing on a rational inattention. Originating in bounded rationality explanations of macroeconomics (Sims, 2003), rational inattention has a simple premise: information may be available, but acquiring and processing it requires attention, and attention is costly (Maćkowiak, 2025, 1). Our contention is that rational inattention is a particularly helpful framework for explaining foreign aid implementation. Bureaucrats at aid agencies and international organizations are career-minded, monitoring is costly, and information is noisy and hard-to-acquire. Consequently, bureaucrats must rationally decide how to best allocate their attention, and some are more effective than others at doing so.

Our argument that the interplay between macro-structural constraints and micro-

[3]See also Briggs (2020) and Ashton et al. (2023).

Table 1: Canonical Theories of Compliance in International Relations and Organizations

| Theory | Core Predictions | Key References |
|---|---|---|
| Enforcement | Monitoring and credible sanctions increase compliance; weak punishment yields delay or violation. | Keohane and Nye (1977), Downs, Rocke and Barsoom (1996), Tallberg (2002) |
| Management | Noncompliance often reflects limited capacity, so providing assistance improves compliance. | Chayes and Chayes (1993) |
| Domestic Politics | Compliance rises when domestic coalitions and institutions favor it and can punish violations. | Dai (2005) |
| Rationalism | States choose precision and flexibility to manage uncertainty. | Koremenos, Lipson and Snidal (2001), Koremenos (2005) |
| Principal-Agent | Compliance depends on agent incentives and principals' ability and willingness to monitor. | Nielson and Tierney (2003), Stone (2011) |
| Constructivism | Compliance increases as norms are internalized and violations become stigmatized via socialization and shaming. | Finnemore and Sikkink (1998), Barnett and Finnemore (2004) |
| Public Choice | Bureaucrats respond to career, budget, and ideology incentives. | Frey (1984), Vaubel (1986) |

founded, individual behavior explains compliance patterns draws from Coleman's (1986) boat, which guides much research in sociology. Following the special issue framework paper (Weaver, Morrison and Heinzel, 2026, 13), we focus on situations of "conforming alignment": that is, pressures that are mutually-reinforcing. We expand these insights by using the rational inattention lens to clarify how structures dominate in these contexts.

## 1.1. Macro-Structural Factors

Before delving into bureaucrats' microfounded behaviors, it is necessary to understand the macro-structural compliance constraints that bureaucrats cannot alter in the short term. Previous work highlights countries' bureaucratic organization, revenue sources (e.g., foreign direct investment and natural resources), and domestic political constituencies (Dai, 2005; Girod and Tobin, 2016; Carcelli, 2024).

We build on these insights by advancing how compliance relates to another structural constraint: geography (Gerring and Christenson, 2017, 65). Geography is not deterministic, as excellent climates are not consistent predictors of development outcomes (e.g., Dell, Jones and Olken, 2012). Instead, geography can drive compliance in areas *within countries* where the state's reach is limited, which increases bureaucratic monitoring costs. In this sense, our contention is related to the enforcement school of compliance as well Mann's (1984) concept of infrastructural power. Incidentally, our rationale for focusing on geography relates to its ability to constrain the state's ability to provide infrastructure and, in turn, foster economic growth and development (Donaldson, 2018; Rogowski et al., 2022).

Significant empirical state capacity research supports the pattern that geographically difficult-to-reach areas produce more development challenges. Müller-Crepon (2023), for example, shows that longer distances to the capital cities in Africa predict lower levels of educational attainment, higher infant mortality, and lower electricity provision. Provenzano (2024) documents a link between these patterns and accountability in Africa, and Campante and Do (2014) showcase their external validity given that they focus on the United States. In the conflict space, Fearon and Laitin (2003) famously argue that rugged terrain fuels civil wars.

All of these empirical patterns related to geography and development support our argument that the interplay of geography and limited state presence makes compliance more difficult. In this way, our argument follows the enforcement, management, and domestic politics schools of compliance described in Table 1 (e.g., Chayes and Chayes, 1993; Downs, Rocke and Barsoom, 1996; Dai, 2005).

## 1.2. Microfounded Bureaucratic Incentives

Macro-structural constraints are not fully deterministic of compliance outcomes, as international bureaucrats and their incentives matter for a wide range of development outcomes (e.g., Poulsen and Aisbett, 2013; Bayram, 2017; Hafner-Burton et al., 2017; Fjelstul and Car-

rubba, 2018; Heinzel, 2022). Principal-agent and public choice theories offer helpful starting points for understanding individual-level factors. They highlight that bureaucrats are self-interested and frequently take into account career, ideology, and pecuniary considerations (e.g., Frey, 1984; Williamson, 1985). Chwieroth (2013, 2015), Nelson (2014), and Lang, Wellner and Kentikelenis (2025) demonstrate empirical support for bureaucrat-related biases at the International Monetary Fund (IMF). Similar phenomena take place at the World Bank, too (Smets, Knack and Molenaers, 2013; Clark and Dolan, 2021).

A salient career-oriented incentive for international bureaucrats working in foreign aid is to satisfy the so-called "disbursement imperative" (e.g., Buntaine, 2016). Essentially, international bureaucrats need to ensure that their projects disburse money, because that is how they advance in their careers. The pressure to complete projects derives from aid agencies' need to loan money for survival: they earn money from interest rates, having a large project portfolio, and AAA bond ratings (Babb, 2009). In turn, some bureaucrats may be incentivized to downplay or overlook some compliance issues. Using a constructivist lens, Weaver (2007) describes these phenomena as "hypocrisy".

At the same time, international bureaucrats self-select into organizations for which they identify with the mission (Frey and Gygi, 1990), so they are at least somewhat mission-driven (Honig, 2018). Additionally, principals monitor bureaucrats and would have engaged in re-contracting if aid agencies did not serve their end goals (Hawkins et al., 2006; Alter, 2008). Such a setting is thus not one where international bureaucrats completely ignore compliance requirements, particularly given meritocratic hiring and constructivists' emphasis on how bureaucratic norms spread (Barnett and Finnemore, 2004). What, then, is the right amount of compliance?

## 1.3. Rational Inattention: Integrating Both Macro-Structural Factors and Bureaucratic Microfoundations

Consistent with the special issue framework paper (Weaver, Morrison and Heinzel, 2026), compliance is not a uniform task. In particular, international bureaucrats' abilities to engender compliance outcomes frequently depend on whether structural pressures are aligning, conflicting, or orthogonal. Our examination of compliance in foreign aid implementation illuminates the case of conforming alignment, where structural and agency pressures reinforce each other. On the structural side, geography and weak state reach combine to form low infrastructural power. On the agency side, international bureaucrats respond to these challenges under the disbursement imperative and the pressure to monitor in a principal-legible way.

In such settings of conforming alignment, we argue that the behavior of international bureaucrats follows a rational inattention framework (Sims, 2003). It clarifies that international bureaucrats not only face imperfect information but also must *choose* how to respond to it given time, cost, and career constraints (Maćkowiak, Matějka and Wiederholt, 2023, 229). Because attention is costly, international bureaucrats rely on cognitive shortcuts and coarse cues. The result is allocation of scarce attention toward problems that appear more tractable, legible to principals, and less "wicked" (see Rittel and Webber, 1973). It is the coarse cue that distinguishes a rational inattention account from one aligning more with standard public choice, principal-agent, or enforcement accounts (see Table 1).

Geography provides for such a coarse cue. International bureaucrats view aid implementation as more difficult in rural locations (Briggs, 2021), but a key reason why is information noise. Under the typical Shannon (1948) information constraints of rational inattention, there is an optimal signal-to-noise ratio. Nevertheless, it requires effort to disentangle. More practically, not every rural area poses the same implementation challenges, but distinguishing among the cases is too costly or difficult for the average international bureaucrat. The consequence is non-linear and tail-heavy effects, where the signal of difficulty is strongest.

Legibility and cost perspectives further illuminate bureaucratic constraints. For example, visiting one geographically remote area may result in the loss of sufficient resources to visit three more accessible areas. That is significant from career-minded bureaucrats' perspectives, because visiting more sites is more countable, thereby affording them more protection in case something goes wrong and principals take punitive action (Scott, 1998; Lee and Zhang, 2017, 118). Following Heinzel, Weaver and Jorgensen (2025), visiting more accessible areas is not only the more "shallow" or "box-ticking" compliance activity, but it is also the safer one. By contrast, deep consultations and remote site visits are less legible and less likely to reach principals, so they incur higher career-related risks for international bureaucrats. For these reasons, international bureaucrats allocate attention in response to infrastructural power, yielding:

**H1: Lower state infrastructural power, as measured by higher travel times, decreases foreign aid compliance.**

It is also useful to distinguish infrastructural power, which involves both geography and state presence, from accounts more centered on geography. In this light, we hypothesize:

**H2: Geographic proximity/density to other foreign aid projects increases compliance.**

Geographic proximity to other projects can improve compliance, as international bureaucrats may have experience working in the same areas and/or can ask their colleagues for assistance. Additionally, geographic proximity reduces the supervision burden by enabling bureaucrats to monitor multiple projects in a single trip. Nevertheless, density measures abstract from road networks and state presence, so density may not capture international bureaucrats' actual supervision feasibility constraints in the same way as infrastructural power.

Finally, rational inattention does not imply equal effects for all bureaucrats. In the context of Shannon (1948) information constraints, some international bureaucrats are able to optimize signal-to-noise acquisition (Sims, 2006), which previous research supports. For

example, Denizer, Kaufmann and Kraay (2013), Bulman, Kolkma and Kraay (2017), Heinzel and Liese (2021), and Denly (2025) all show that higher-performing international bureaucrats improve various types of aid outcomes. Along these lines, we hypothesize:

> **H3: Higher-performing international bureaucrats are able to mitigate structural development constraints.**

Consistent with our pre-analysis plan, we use the word "mitigate", not overcome. International bureaucrats lack sovereignty to operate in aid-receiving countries (Nielson and Tierney, 2005), and recipient countries are formally responsible for most implementation, particularly following the 2005 Paris Declaration on Aid Effectiveness (OECD, 2005). There is thus only so much bureaucrats can achieve, particularly in the face of structural constraints.

# 2. Research Design

## 2.1. Pre-Analysis Plan

Our anonymous pre-registration prior to running any analysis is available at OSF.[4] We do not deviate from the pre-analysis plan.

## 2.2. Setting

We test the relevance of our argument using data from the world's largest and most influential international development organization, the World Bank. According to AidData (Tierney et al., 2011), from 1947-2013 the World Bank financed 42% of multilateral foreign aid commitments, roughly accounting for $US2013 1.66 trillion (Denly, 2021). The World Bank offers numerous services, including loans, credits, guarantees, data, and technical assistance (Kramarz and Momani, 2017).

We examine projects financed by both the World Bank's concessional lending arm, the International Development Association (IDA), and the institution's market-based lend-

[4]See: https://osf.io/ru9dp/?view_only=e7458ea391bd4a90b7ffae2b5baba451.

ing arm, the International Bank for Reconstruction and Development (IBRD). Specifically, we focus on investment project financing (IPF),[5] which provides input-based financing for specific investments, such as in infrastructure, health, governance, etc. We do not consider structural adjustment/development policy loans, because they disburse on the basis of completing prior actions, not monitorable inputs (World Bank, 2012). We also exclude Program-for-Results (PforR) loans, as they only focus on recipient governments' existing programs and require different monitoring requirements than a typical development project.

The World Bank executive board formally approves and oversees all projects and operational initiatives, but Task Team Leaders (TTLs) are responsible for project management. That mainly includes project supervision, not implementation, which is generally the responsibility of aid-receiving countries.[6] There are several layers of directors and managers above TTLs, but TTLs' daily work is substantial. Beyond supervising project implementation and site visits, TTLs conceive of projects in line with Country Partnership Frameworks, consult with recipient country stakeholders and bureaucrats, and shepherd projects through executive board approval (Heinzel and Liese, 2021, 628-629). TTLs also manage other staff members working on the three main buraucratic hassle factors: loan processing, procurement, and environmental and social safeguards (Humphrey, 2015).

The empirical analysis in this paper focuses on the third hassle factor, environmental and social safeguard compliance. Generally, the World Bank has the most stringent safeguard requirements (Humphrey, 2015; Greenstein, 2022). A lot of these requirements are a result of severe failures, which exposed the World Bank to legitimacy challenges and generated significant attention from principals (Nielson and Tierney, 2003; Park, 2022). Buntaine (2016) demonstrates the legacy of these failures, finding that borrowing countries with safeguard failures receive lower aid allocations in future periods. Bureaucrats need to complete projects to advance their careers, but safeguard challenges are difficult to solve and failures

[5]Previously, IPF was called Investment Lending.

[6]Especially following the 2005 Paris Declaration on Aid Effectiveness (OECD, 2005), aid-receiving countries are responsible for project implementation.

slow down implementation, yielding a chilling effect on project approvals.

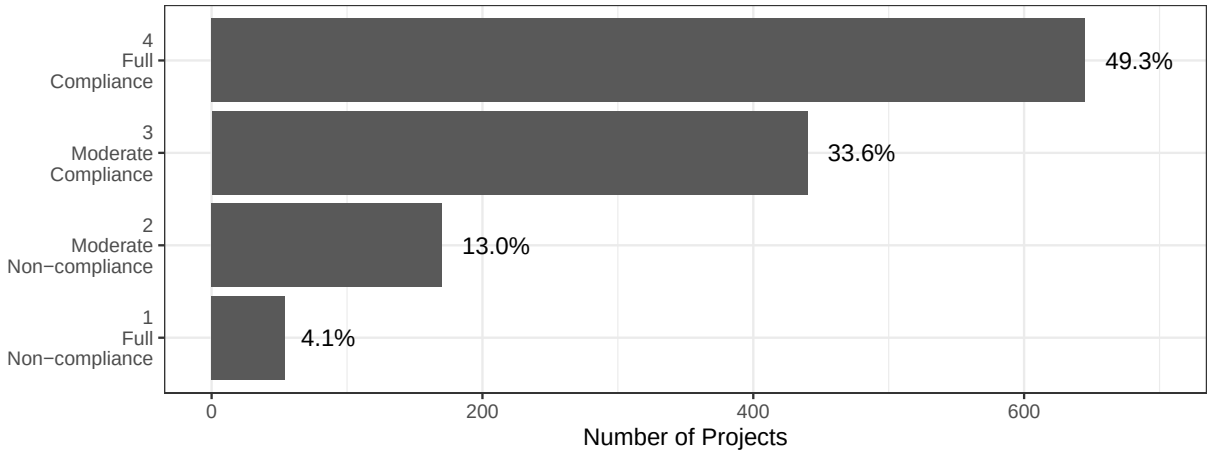
Ethnographic analyses of safeguard compliance confirm the difficulty of World Bank bureaucrats' practical constraints, relating to cost and information. Tello (2015), for example, discusses problems such as indigenous peoples lacking property titles, corruption in their allocation, and settlers burning down indigenous property. Without time and resources, bureaucrats do not have sufficient information to solve these challenging problems and prevent people from the negative externalities of aid. Randeria and Grunder's (2011) analysis of resettlement in the context of the World Bank's Mumbai Urban Transport Project provides another relevant example. In this instance, World Bank bureaucrats could not gather sufficient information and support the 120,000 people affected by resettlement. These tragedies happened under poor living conditions, the government ignoring citizens' legally-defensible grievances, and numerous other problems that brought real harm to citizens. In some sense, the compliance problems that both Randeria and Grunder (2011) and Tello (2015) expose are even more "unimplementable by design" than the ones Reinsberg, Stubbs and Kentikelenis (2022) highlight at the IMF.

## 2.3. Dependent Variable

The primary outcome variable is whether each respective World Bank project complies with the institution's social and environmental safeguard policies. These policies concern environmental assessment; involuntary resettlement; physical cultural resources; indigenous peoples; natural habitats; pest management; forests; international waterways; dams; disputed areas; and environmental action plans. We take the variable from Denly (2025), whose coding covers non-supplementary IPFs that are subject to safeguard policies for the approval years 2007-2015.[7] Denly (2025) chooses this time period to capture the years in which World Bank safeguard-related guidance are subject to official policies, not mere directives, prior to the 2016 change in safeguard policies (see Greenstein, 2022). Similar to Buntaine (2016),

[7]Supplementary/additional financing projects are not subject to safeguard policy review, which is why Denly (2025) excludes them.

Figure 1: Summary Statistics of World Bank Safeguard Policy Compliance (2007-2015)




Source: Denly (2025)

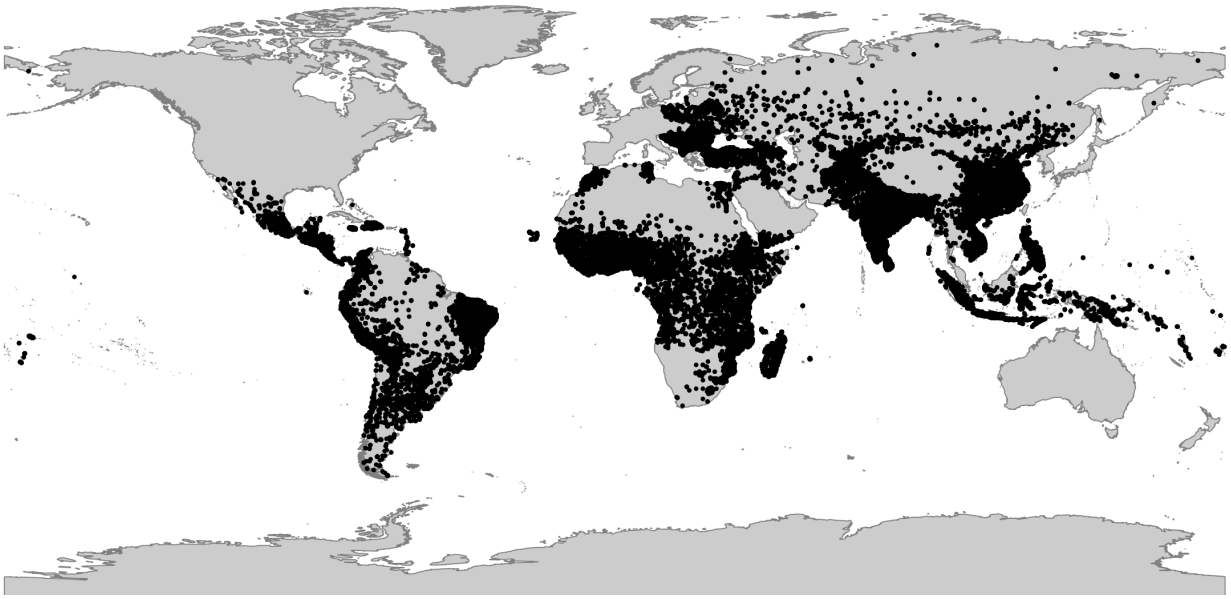
Denly (2025) codes safeguard compliance on a 1-4 scale, in which 4 represents full compliance; 3 represents moderate compliance; 2 represents moderate non-compliance; and 1 represents full non-compliance. Overall, the data correspond to the safeguard policy compliance outcomes for 1,309 projects approved from 2007-2015. Figure 1 presents relevant summary statistics. They show that around half (49%) of the projects are in full compliance; 34% of projects enjoy moderate compliance; and 17% of projects exhibit some form of moderate or full non-compliance.

## 2.4. Treatments

### 2.4.1. Travel Time

Our main treatment variable to capture the effect of geography and state presence is the natural log of travel times to access World Bank project locations. We draw our baseline spatial World Bank project data from Bomprezzi et al.'s (2025) Geocoded Official Development Assistance Dataset (GODAD). In cases where GODAD does not include the respective latitudes and longitudes of each project, we take them from the World Bank's external website as well as a freedom of information request. In total, we are able to obtain location data for all but 21 of the 1,309 projects in Denly's (2025) dataset, corresponding to

Figure 2: Locations of World Bank Projects



Sources: Bomprezzi et al. (2025) and the World Bank.

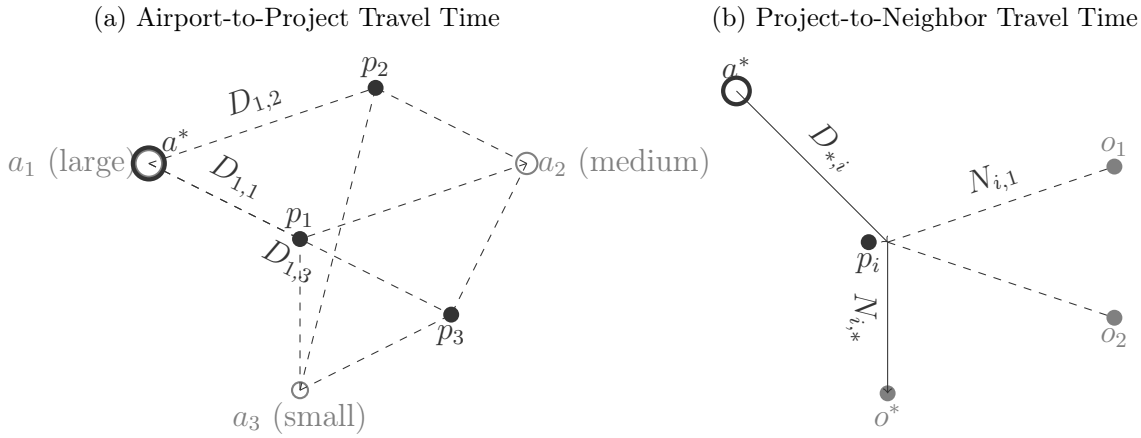
a missingness rate of 1.6%. Figure 2 provides a map of all projects in the dataset.

For the travel time calculations, we partition the Open-Source Routing Machine into regional servers using the OpenStreetMap Foundation's (2025) maps in line with Appendix G. Our calculations account for the actual shapes of the roads, stops, speed limits, and traffic. Three algorithmic steps that we detail in Appendix F, Figure 3, and below underpin our calculations.

In Step 1, we use geolocated data from Megginson (2025) on 48,000 worldwide airports to measure the closest, appropriately-sized airport in minutes to each project's most accessible location. In doing so, we direct the algorithm to search within each country of the project and prioritize large airports over medium-sized ones and medium-sized airports over small-sized ones. The rationale behind the prioritization is that international bureaucrats usually fly into a larger airport, assuming one exists.

In Step 2, we calculate the driving time in minutes to the nearest location of another project. It differs on a yearly basis as new projects start and close, which allows for the data

Figure 3: Airport Selection and Travel Time Algorithms




Note: The above figure provides a graphical illustration of the combined optimal travel time calculations. As we detail in Appendix F.2, we first use an airport selection algorithm. It evaluates project locations $p_1, p_2, p_3$ (filled circles) against airports $a_1$ (large), $a_2$ (medium), and $a_3$ (small). Then, the algorithm sums the travel times, $D_{j,i}$, to obtain the total travel time for each airport, $S_j$. Finally, the algorithm selects $a^*$ if $S_j > 0$ and is the lowest. Then, per Appendix F.3, we calculate the nearest neighbor project location, $o^*$, by minimizing the project-to-project travel time in seconds, $N_{i,*}$, from the closest other project in the same country, $o$, among all of its locations $o_1, o_2, o_3$. The total travel time is $D_{*,i} + N_{i,*}$, for which we take the log to minimize the influence of outliers. Although we present the above figure using straight lines for readability purposes, our actual routes follow the actual shapes of the roads and take into account stops, speed limits, and traffic patterns. Appendix F provides the full mathematical presentation of the calculations.

to be time-varying. Overall, 38% of the project-specific driving times change each year, and the average change in driving time is 8 minutes. As with the airport-to-project calculations, the project-to-neighboring project calculations are country-specific. In other words, these calculations do not allow for the crossing of country boundaries, even if the route is faster. We adopt such a strategy because TTLs rarely supervise projects from different countries in *one supervision mission* without taking a different flight. It is also generally not straightforward to cross international borders in the low- and middle-income countries that receive World Bank projects.

In Step 3, we sum the minute-based driving times from Step 1 and Step 2. Then, we take the natural log to capture proportional increases in monitoring burden and reduce the leverage of extremely remote observations. Using the log specification lets the variable capture proportional changes in monitoring burden, which fits our rational inattention argument.

Table 2 provides relevant summary statistics. We supplement the overall statistics with country-level ones for Bangladesh, Ethiopia, and Kazakhstan to provide a plausibility check for the travel-time measure. Ostensibly, Bangladesh is a much smaller country than Kazakhstan, so it is intuitive that average driving times are shorter in Bangladesh. Similarly, the size of Ethiopia is somewhere between the sizes of Bangladesh and Kazakhstan, making it logical that Ethiopia's summary statistics fall in the middle. With respect to large maximum values for the overall panel, they often pertain to projects in the Brazilian Amazon, where the road network is extremely sparse. In any event, we are not concerned by these large values. Even if TTL supervision missions to such places normally involve taking a small plane or helicopter to an airport that is outside of Megginson's (2025) airport data, it remains that such a location is very difficult or costly to access and supervise. Furthermore, because our final travel time variable incorporates a natural logarithm, we minimize the influence of outliers in very remote locations and enable proportional, non-linear interpretation, consistent with our rational inattention theory. Again, we pre-registered this specification, and log driving times in Table 2 follow a respectable range.

#### 2.4.2. TTL-Specific Travel Time

Consistent with our pre-analysis plan, we also calculate TTL-specific travel times that distinguish by the name of the project Task Team Leader (TTL) in each country-year, using data from Denly (2025). The data come from the name of the TTLs at each World Bank project's Implementation Status Reports (ISRs). In the rare cases where TTLs switch within a given calendar year, we take the first chronological TTL in the list.[8]

Despite our efforts to avoid missingness, it is impossible to avoid it for these TTL-specific variables. The reason is simple: TTLs do not always supervise more than one project in a respective country at the same time. In total, we are only able to capture 17%

[8]Given that Denly's (2025) data only extend through 2021, we fill down the names of the TTL for projects that are still running in 2022 and 2023. Such instances are also rare: Denly's (2025) safeguard compliance data refer to projects approved from 2007 to 2015, and there are very few projects still running in 2022 or 2023, as World Bank projects take around 5 years to complete.

Table 2: Raw Driving Time Summary Statistics

| Variable | N | Mean | SD | Min | Max |
|---|---|---|---|---|---|
| Panel A: Overall | | | | | |
| Airport to Location | 368,385 | 410.81 | 568.55 | 0.00 | 18754.20 |
| Location to Nearest Neighbor | 379,462 | 71.77 | 280.55 | 0.00 | 16961.20 |
| Total Travel Time | 366,990 | 481.98 | 758.43 | 0.00 | 35715.40 |
| Airport to Location (Log) | 368,385 | 5.44 | 1.16 | 0.00 | 9.84 |
| Location to Nearest Neighbor (Log) | 379,462 | 3.51 | 1.23 | 0.00 | 9.74 |
| Total Travel Time (Log) | 366,990 | 5.65 | 1.08 | 0.00 | 10.48 |
| Panel B: Bangladesh | | | | | |
| Airport to Location | 17,152 | 167.59 | 84.62 | 12.10 | 627.80 |
| Location to Nearest Neighbor | 17,152 | 18.37 | 23.93 | 0.00 | 464.90 |
| Total Travel Time | 17,152 | 185.96 | 95.98 | 16.70 | 1092.70 |
| Airport to Location (Log) | 17,152 | 4.97 | 0.62 | 2.57 | 6.44 |
| Location to Nearest Neighbor (Log) | 17,152 | 2.61 | 0.90 | 0.00 | 6.14 |
| Total Travel Time (Log) | 17,152 | 5.09 | 0.57 | 2.87 | 7.00 |
| Panel C: Ethiopia | | | | | |
| Airport to Location | 7,204 | 414.84 | 233.46 | 1.80 | 1486.10 |
| Location to Nearest Neighbor | 7,204 | 67.37 | 65.25 | 0.00 | 741.20 |
| Total Travel Time | 7,204 | 482.21 | 265.33 | 9.60 | 2227.30 |
| Airport to Location (Log) | 7,204 | 5.73 | 1.00 | 1.03 | 7.30 |
| Location to Nearest Neighbor (Log) | 7,204 | 3.76 | 1.13 | 0.00 | 6.61 |
| Total Travel Time (Log) | 7,204 | 5.91 | 0.95 | 2.36 | 7.71 |
| Panel D: Kazakhstan | | | | | |
| Airport to Location | 1,750 | 823.05 | 616.59 | 1.20 | 2907.00 |
| Location to Nearest Neighbor | 1,750 | 158.50 | 174.21 | 0.00 | 2069.00 |
| Total Travel Time | 1,750 | 981.55 | 705.30 | 29.60 | 4976.00 |
| Airport to Location (Log) | 1,750 | 6.25 | 1.19 | 0.79 | 7.98 |
| Location to Nearest Neighbor (Log) | 1,750 | 4.37 | 1.58 | 0.00 | 7.64 |
| Total Travel Time (Log) | 1,750 | 6.50 | 1.09 | 3.42 | 8.51 |

Note: The unit of analysis for these summary statistics is the project-location-year. Driving times are measured in minutes. See Appendix F for the full mathematical details behind our algorithmically-derived calculations.

of the 180,000 country-location-years with the TTL-specific variables. Given this low share, we use both standardized mean differences (SMDs) and p-values to analyze the balance of all pre-treatment covariates in Table A.3. We observe that none of the covariates are balanced, as p-values are all $< 0.001$ and SMDs all exceed $\pm$ 0.5. Against this backdrop, it is clear that the TTL-specific data are not a random sample of all projects and likely pose

selection problems that make inference tenuous. Consequently, while we still conduct the relevant analyses to be consistent with our pre-analysis plan, we only report these analyses in Appendix E.

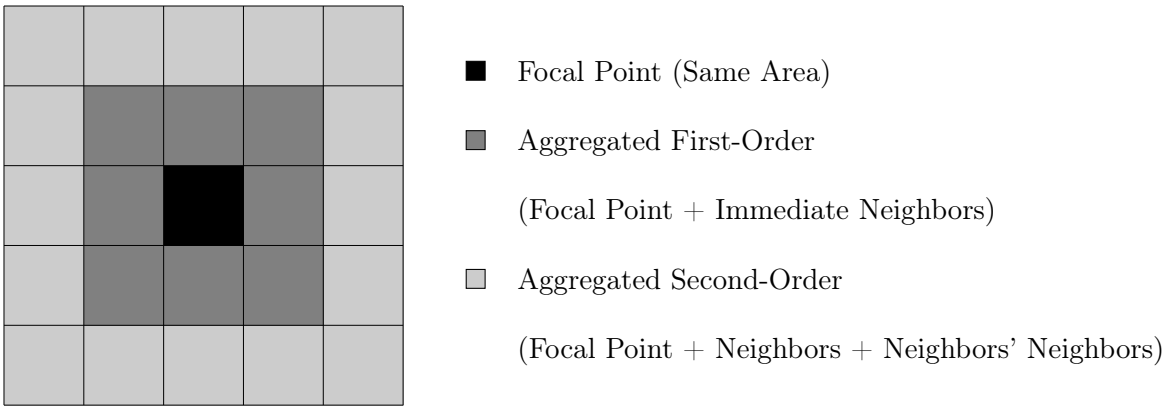
### 2.4.3. Spatial Density

As our primary alternative to the overall travel time treatment that captures geography and institutions, we calculate the density of projects operating within the same areas as well as neighboring ones. Compared with the driving time variable, the density variable is more related to geography. To operationalize the same versus neighboring areas, we use the PRIO GRID to aggregate the project locations data into 0.5 degrees longitude by 0.5 degrees latitude grid cells (Tollefsen, Strand and Buhaug, 2012). Roughly, these grid cells correspond to $55 \times 55$ kilometer grid cells at the equator, and they become smaller at higher latitudes. Specifically, we calculate the number of unique projects operating within:

1. the same grid cell (focal point);
2. all immediately neighboring grid cells (first-order spatial lag);
3. all of the neighbors' neighboring grid cells (second-order spatial lag).

Given our focus on how project density increases with area, we aggregate the above quantities using the Queen Contiguity method (Darmofal, 2015, 15-17), which we graphically illustrate in Figure 4. In turn, the aggregated first-order spatial lag category also includes the number of projects in the focal point. Similarly, the aggregated second-order spatial lag includes the number of projects in both the focal point grid cell and first-order neighbors.

Figure 4: Aggregated Spatial Lag Diagram (Queen Contiguity Method)




## 2.5. Moderating Variable for the Interactions

For all the interactions to capture whether bureaucratic agency can moderate geographic pressures, we use Denly's (2025) measure of TTL performance for each project. Denly's (2025) measure attempts to mimic earlier coding from Denizer, Kaufmann and Kraay (2013), Bulman, Kolkma and Kraay (2017), and Ashton et al. (2023) using publicly-available data. Specifically, Denly (2025) codes (i) the TTL name at each project's bi-annual Implementation Status Report (ISR); (ii) computes each TTL's daily average project Independent Evaluation Group (IEG) outcome score for his/her completed projects, excluding the current and future ones to prevent circularity; (iii) merges the daily average IEG scores back into each ISR for each project; and (iv) averages across all ISRs for each respective project to obtain a weighted TTL performance score that often reflects multiple TTLs.

## 2.6. Control Variables

### 2.6.1. Project-Level Variables

Given that larger projects may require more attention from TTLs, we control for the commitment amount (i.e., project loan/credit size). These commitment amounts involve millions and sometimes billions of US dollars, so we deflate the data to constant dollars and take the natural log. Additionally, we control for the safeguard category, which captures

the pre-project-approval safeguard risk. It makes sense to control for it given that projects with higher risks may prompt TTLs to dedicate more time to their supervision. We do not control for supervision or project costs because doing so would invoke post-treatment bias.

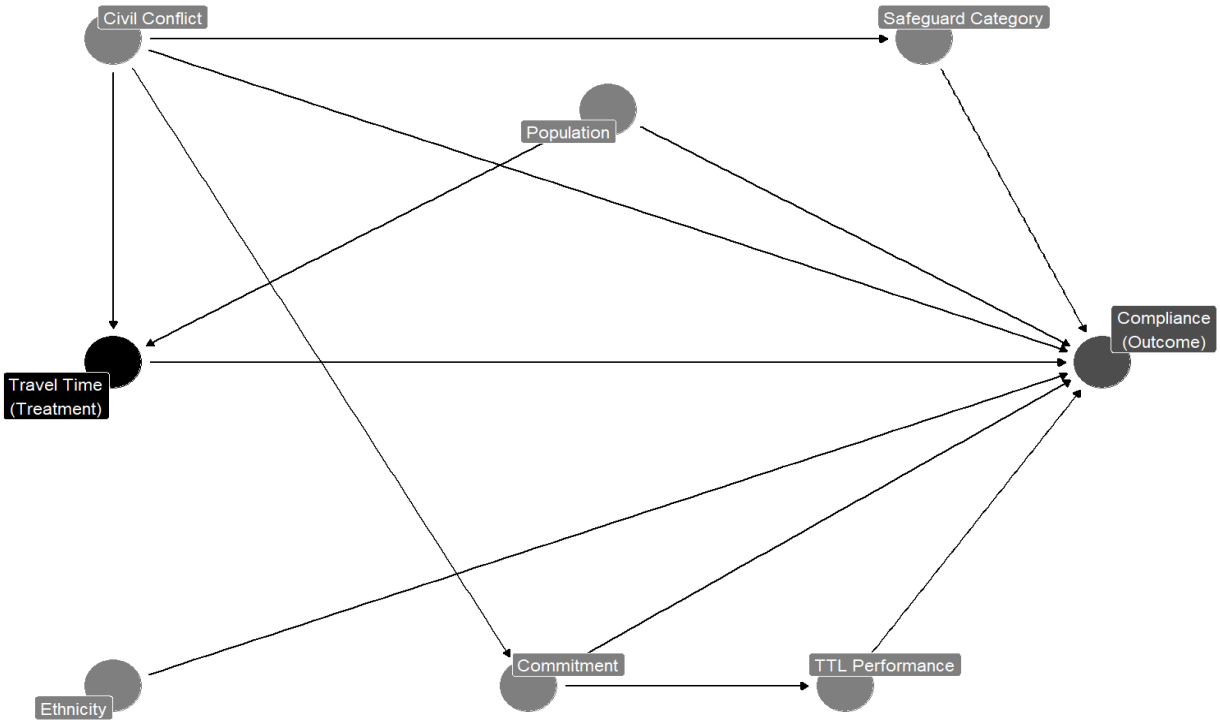
### 2.6.2. Spatial Control Variables

A particularly essential control variable is the spatial measure of log population counts per square kilometer from the HYDE project (Klein Goldewijk, Beusen and Janssen, 2010; Goldewijk et al., 2017).[9] The measure is essential because failing to control for it would make it difficult to determine whether the results reflect population patterns or the geography/state-reach mechanism captured by travel time and density. By controlling for the variable, we explicitly avoid difficult-to-adjudicate debates on the role of population (e.g., Herbst, 2000; Robinson, 2002; Bandyopadhyay and Green, 2012). We can be confident about that conclusion because, unlike other social science studies using HYDE data aggregated by the PRIO GRID, we do not use five-year intervals and linear interpolation to fill in missing values. Instead, we rasterize the original HYDE data to obtain full panels of yearly values.

We also control for two additional spatial controls. To account for ethnicity, we use the measure of excluded ethnic groups from Vogt et al. (2015). Accounting for ethnicity is essential in the context of safeguard compliance: politically marginalized ethnic groups, including many indigenous peoples, must frequently resettle and lose culturally-relevant property as a result of World Bank projects. We considered using Denly et al.'s (2022) natural resource measure, but it contained too much missingness to include. Finally, we use UCDP's spatial conflict data from Pettersson, Högbladh and Öberg (2019). It accounts for the effect that conflict may have on the ability to travel to an area.

[9]The pre-analysis plan also specified that we would control for the (log) population counts per square kilometer; (log) population density per square kilometer; (log) urban population counts; (log) rural population counts; and (log) built-up area. In reality, all of these variables are extremely highly correlated (see Figure A.2). Additionally, most of social science uses the HYDE population count variable, so that is what we use here for the main analyses. Figure C.2 provides sensitivity analyses that individually control for the different population measures.

Figure 5: DAG for Determining the Adjustment Sets




Note: See Appendix B for a full verbal description of all paths.

## 2.7. Identification and Determining the Adjustment Sets

To improve targeting of the average treatment effect, we draw a Directed Acyclic Graph (DAG) to determine our final adjustment sets. Figure 5 depicts the DAG, which suggests that there are no mediators, colliders, or descendants that alter the target estimand.

For the models focusing on the effect of travel time (or density) on compliance, civil conflict and population variables comprise the minimal adjustment set: they are the only two variables that directly affect both the treatment and the outcome (see Pearl, 2009). The rest of the independent variables enter as precision-oriented controls in our canonical adjustment set specifications (see Perković et al., 2018).

The models focusing on the interactive effect of TTL performance follow a similar logic. The only variable that directly affects both TTL performance and compliance is the commitment amount, so it constitutes the minimal adjustment set. The other independent variables

take the form of precision-oriented controls for the canonical adjustment set estimates.

Appendix B elucidates that most of the non-outcome relationships that the DAG showcases are straightforward, but it is worth further probing the relationship between TTL performance and the safeguard risk category. The World Bank may wish to protect its reputation by assigning better TTLs to riskier projects, suggesting possible endogeneity concerns, but the data do not reflect such patterns. Figure A.1 indicates that the pairwise correlation between TTL performance and the safeguard risk category is merely 0.12. To further guard against endogeneity risk, Figure A.1 also considers the overall Independent Evaluation Group (IEG) project outcome scores and finds that they correlate with the safeguard risk categories at 0.11. In short, available data and DAG suggest that endogeneity represents a low risk for these models.

## 2.8. Final Unit of Analysis and Estimand

As the previous sections demonstrate, the main variables of interest operate at different units of analysis. Notably, compliance is a project-level measure, but travel time and density vary across locations and years. New projects opening and others closing are responsible for these dynamics.

To address these challenges, we follow our pre-analysis plan and use the project-cell-year as the unit of analysis. We construct the project-cell-year dataset by creating separate rows for each year in which each project location is active, starting with the project approval year and ending with the project closing year. After calculating location-specific travel times (see Section 2.4.1), we use the PRIO GRID to aggregate project locations into roughly 55 $\times$ 55 kilometer grid cells, corresponding to 0.5 degrees latitude by 0.5 degrees longitude (Tollefsen, Strand and Buhaug, 2012). After aggregating, the dataset contains 105,689 project-cell-years.

To respect the fact that the outcome is at the project level, we do not treat project-cell-years as separate compliance outcomes. Instead, project-cell-years represent the exposure

environments attached to each project over space and time. To prevent projects with more locations or longer implementation periods from dominating the analysis, we inversely weight the data by each project's number of project-cell-years. Each project thus contributes the same total weight to the estimating equation, while the treatment and spatial controls retain their project-cell-year variation.

The resulting project-cell-year dataset offers numerous benefits. First, using grid cells enables us to capture projects located in similar areas with a consistent aggregator. Second, the project-cell-year dataset enables us to capture within-project variation in local conditions for the independent variable. Third, it allows us to merge in numerous spatially-relevant control variables that are not available at the spatial point level but are available at the raster level (see Section 2.6.2).

To be clear, the project-cell-year is not the only possible aggregation. The most plausible alternative would aggregate by the project-level mean, but that would build a static exposure rule into the main estimand, which is clearly dynamic (see Step 2). Additionally, if there is any non-systematic measurement error, which is possible given that OpenStreetMaps does not know about every road in the world, it would bias estimates toward zero (King, Keohane and Verba, 1994, 163–168). The project-cell-year unit of analysis thus has many more attractive properties than one aggregated to the project level.

## 2.9. Statistical Models

Consistent with Berman et al. (2017) and Denly et al. (2022), we use a spatial heteroskedastic and autocorrelation consistent (HAC) linear regression model:

$$\text{compliance}_{(k,i,t)} = \beta_{\text{log travel time/density}_{(k,i,t)}} + \beta_{\text{controls}_{(k,i,t)}} + FE + \epsilon_{(k,i,t)} \tag{1}$$

The interaction models also follow a similar structure:

$$\begin{aligned}\text{compliance}_{(k,i,t)} =& \beta_{\text{log travel time/density}_{(k,i,t)}} + \beta_{\text{TTL Performance}_{(k,i,t)}} + \\ & \beta_{\text{log travel time/density}\times\text{ TTL Performance}_{(k,i,t)}} + \beta_{\text{controls}_{(k,i,t)}} + FE + \epsilon_{(k,i,t)}\end{aligned} \tag{2}$$

where subscript $k$ refers to the respective grid cell, $i$ corresponds to the project, $t$ denotes the time/year, $\epsilon$ is an error term, and $FE$ corresponds to fixed effects. With respect to the latter, we consider fixed effects at the country and year levels.

Within-grid-cell effects are irrelevant for the question and estimand of interest, so we do not include grid cell fixed effects. Fundamentally, the study concerns whether greater travel times or geographic project densities reduce compliance. Because grid cell fixed effects would pull interpretation back within the same grid cell, the resulting estimand would become difficult to interpret and irrelevant for the research question.

We do not include project fixed effects for similar reasons. Project fixed effects would be useful if the-airport-to-project variation (Step 1) was ignorable. However, it is essential for understanding monitoring burdens of international bureaucrats.

Given our focus on spatial driving time and densities, the most appropriate standard error specification entails the use of Conley (1999) standard errors. To account for spatial dependence, we specify distance cutoffs for potential spatial correlation of 150 kilometers, thereby capturing grid cells within the second to third order spatial lag of each other (see Figure 4). As a robustness measure, we also consider spatial cutoffs of 75 kilometers not only because it is half the 150 threshold, but also because the mean project-to-neighbor distance is 82 kilometers. Using such a structure with Conley standard errors has two key benefits. First, Conley standard errors account for the spatial nature of our estimations by adjusting the covariance matrix so that inference is more reliable when nearby observations have correlated errors. Second, using Conley standard errors avoids ambiguous decisions concerning whether to cluster at the level(s) of the grid cell, project, country, year, or the

15 combinations thereof.[10]

# 3. Results

## 3.1. Travel Time Results

Columns (1)-(4) of Table 3 present the main results for the travel time treatment, using a spatial cutoff of 150 kilometers. Recall that travel time enters the regressions in natural log form to correspond with our theoretical focus on rational inattention from bureaucrats. Consequently, we use the summary statistics in Table A.4 to translate the canonical adjustment set estimates in Columns (3)-(4) of Table 3 into changes over the observed distribution of travel times.

Results with both country fixed effects as well as country and year fixed effects are nearly identical in terms of magnitude and explained variance. All estimates are highly statistically significant, as $p = 0.02$ in the minimal adjustment set and $p = 0.006$ in the canonical adjustment set. In terms of the coefficient magnitudes, we begin with the interquartile range (IQR). The IQR estimates suggest that moving from the 25th percentile (199 minutes) to the 75th percentile (657 minutes) of travel time decreases compliance by 0.046 points on the 1–4 scale.[11] That corresponds to a 1.5% decrease in the scale range.[12]

[10] $4^2 - 1 = 15$

[11] $\Delta y = -0.039 \times 1.19 = -0.046$. Note that $6.49 - 5.30 = 1.19$, which represents the difference between 25th and 75th percentile values on the log scale according to Table A.4.

[12] The scale range decrease is calculated as $0.046/3 = 0.015$, given that compliance outcomes are on a 1-4 scale.

Table 3: Travel Time and the Interactive Effect of TTL Performance on Compliance (150 KM Spatial Cutoff)

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| Total Travel Time (Log) | −0.032** | −0.032** | −0.039*** | −0.039*** | −0.208*** | −0.208*** | −0.195** | −0.196** |
| | (0.014) | (0.014) | (0.014) | (0.014) | (0.080) | (0.080) | (0.088) | (0.088) |
| Travel Time × TTL (Log) | | | | | 0.044** | 0.044** | 0.038* | 0.039* |
| | | | | | (0.019) | (0.019) | (0.021) | (0.021) |
| TTL | | | | | 0.015 | 0.015 | 0.018 | 0.017 |
| | | | | | (0.117) | (0.117) | (0.129) | (0.129) |
| Commitment (Log) | | | −0.063*** | −0.066*** | −0.158*** | −0.158*** | −0.119*** | −0.117*** |
| | | | (0.021) | (0.021) | (0.023) | (0.023) | (0.023) | (0.023) |
| Safeguard Category | | | 0.245*** | 0.244*** | | | 0.236*** | 0.236*** |
| | | | (0.040) | (0.040) | | | (0.044) | (0.044) |
| Conflict Count | 0.023 | 0.022 | 0.016 | 0.015 | | | 0.000 | 0.003 |
| | (0.020) | (0.020) | (0.020) | (0.020) | | | (0.023) | (0.023) |
| Ethnic Groups | | | 0.008 | 0.008 | | | 0.010 | 0.012 |
| | | | (0.016) | (0.016) | | | (0.017) | (0.017) |
| Population (Log) | −0.009 | −0.009 | −0.009 | −0.009 | | | −0.010 | −0.009 |
| | (0.009) | (0.009) | (0.010) | (0.010) | | | (0.010) | (0.010) |
| Observations | 101 264 | 101 264 | 97 308 | 97 308 | 89 276 | 89 276 | 85 180 | 85 180 |
| $R^2$ | 0.190 | 0.191 | 0.212 | 0.214 | 0.241 | 0.243 | 0.254 | 0.255 |
| $Adj.R^2$ | 0.189 | 0.190 | 0.211 | 0.212 | 0.240 | 0.241 | 0.252 | 0.254 |
| Country FE | X | | X | | X | | X | |
| Country + Year FE | | X | | X | | X | | X |

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; spatial cutoff at 150 kilometers; minimal adjustment sets in (1), (2), (5), and (6); canonical adjustment sets in (3), (4), (7), and (8)

Outside the IQR, the distribution is highly skewed and changes in geographic accessibility are multiplicative. For example, moving from the median (372 minutes) to the 95th percentile (1542 minutes) results in a decrease in 0.055 compliance points, corresponding to a 1.8% decrease in the scale range.[13] The jump from 1.5% in the IQR to 1.8% in the median-95th percentile scale range, despite capturing 5% less of the distribution, implies a 20% decline in compliance.[14] If we instead investigate the median-99th percentile of driving times (372-3178 minutes), thereby still excluding extreme outliers, there is a 2.8% decrease in compliance, corresponding to a 87.6% decrease with respect to the IQR.[15]

Overall, the above patterns show that geographic structures in the form of travel times have modestly negative impacts on safeguard compliance in the middle of the distribution and much larger impacts toward the upper tail. These patterns fit rational inattention theory: when career-minded bureaucrats have cost, time, and information constraints, geographic structures provide an informative but coarse cue, as clear signals primarily come from the tail of the distribution.

## 3.2. Interaction of Travel Time and TTL Performance

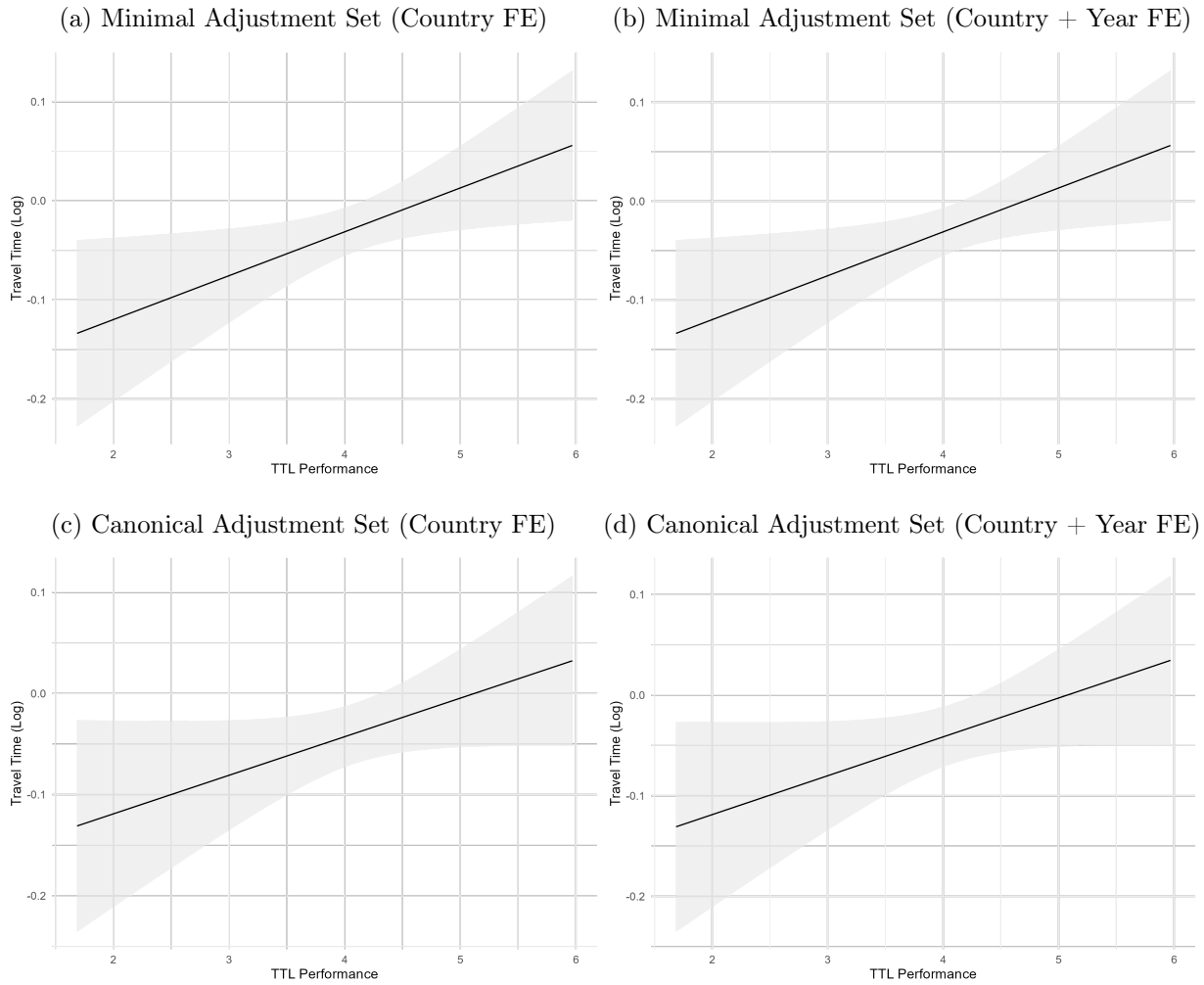
Columns (5)-(8) of Table 3 present the main results for the interaction of travel time and TTL performance, again using a spatial cutoff of 150 kilometers. As with the travel time main effect estimates, results with both country fixed effects as well as country and year fixed effects are essentially identical in terms of magnitude and variance explained. Minimal adjustment set estimates in Columns (5)-(6) are statistically significant ($p = 0.02$) and suggest that higher-performing TTLs can mitigate the structural constraints imposed by travel times. The canonical adjustment set estimates in Columns (7)-(8) are also posi-

[13] $\Delta y = -0.039 \times 1.42 = -0.055$. Note that $7.34 - 5.92 = 1.42$, which represents the difference between the log values at the 50th and 95th percentiles according to Table A.4. The scale range decrease is calculated as $0.055/3 = 0.018$

[14] $(1.5 - 1.8)/1.5 = -20\%$

[15] $\Delta y = -0.039 \times 2.14 = -0.083$. Note that $8.06 - 5.92 = 2.14$, which represents the difference between the log values at the 50th and 99th percentiles according to Table A.4. The scale range decrease is calculated as $0.083/3 = 0.028$, and the percent change with respect to the IQR is calculated as$(1.5 - 2.8)/1.5 = -87.6\%$.

Figure 6: Marginal Effects for the Travel Time × TTL Interactions (150 KM Spatial Cutoff)




Note: The above estimates reflect 95% confidence intervals.

tive but only marginally statistically significant ($p = 0.07$). By extension, these estimates are suggestive that higher-performing TTLs can partially address—but not overcome—the structural constraints imposed by travel times.

The marginal effect plots in Figure 6 help with better understanding the respective effect sizes. Consistent with Table 3, the point estimates meaningfully increase across the different levels of TTL performance. Until reaching an average TTL performance score of around 4.2/6, the effect of travel time on safeguard compliance remains negative and statistically significant. At an average TTL performance score of around 5/6, the conditional marginal

effect of travel time becomes positive. While the coefficient continues to increase monotonically as it approaches the perfect TTL performance rating of 6/6, the lower confidence intervals never eclipse zero for both the minimal and canonical adjustment sets. Overall, international bureaucrats' abilities to address structural impediments to compliance from longer travel times are positive. However, even the highest-performing international bureaucrats cannot fully overcome domestic structural constraints.

### 3.3. Density Results

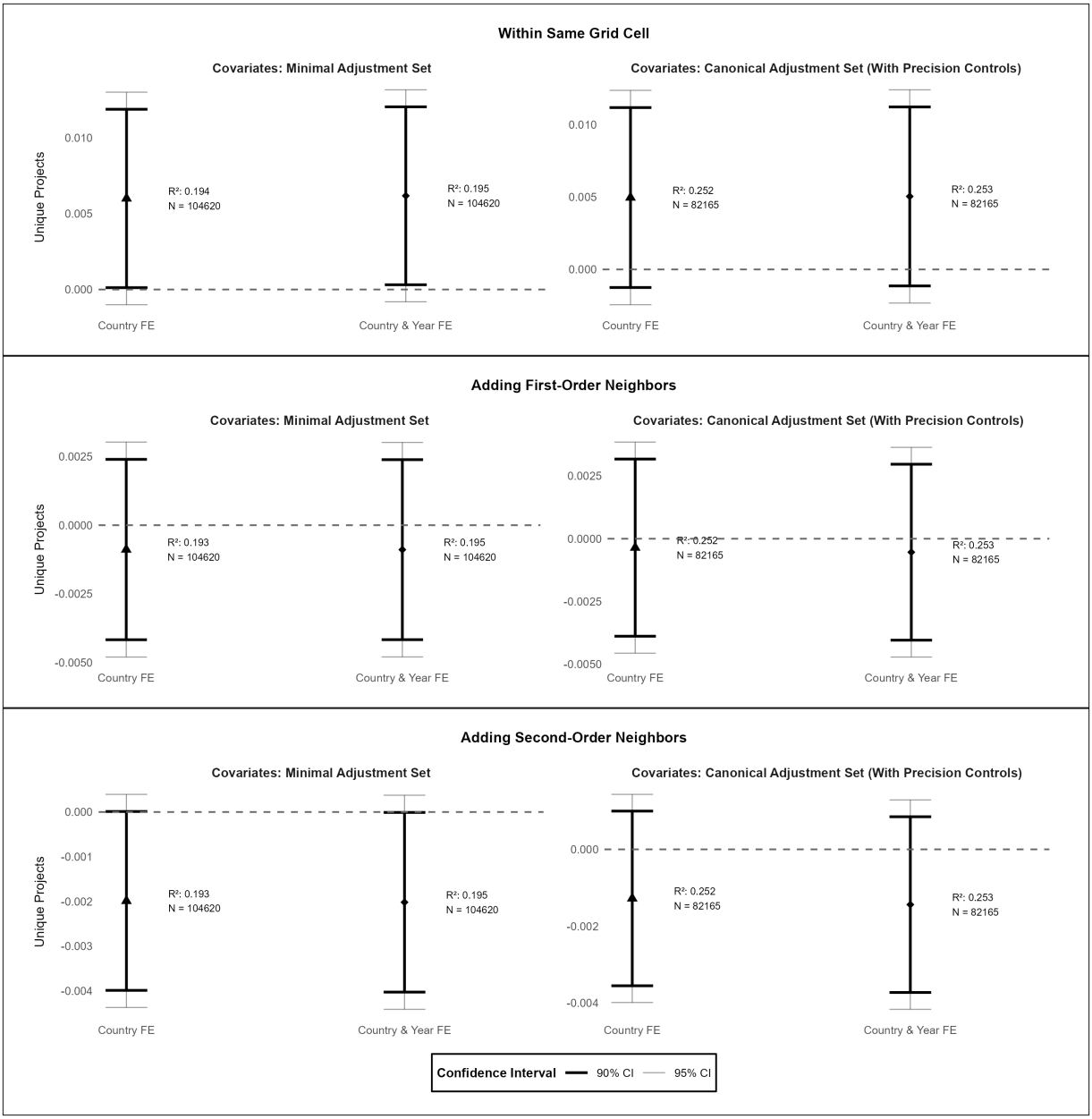
Figure 7 presents the main results of our spatial density analysis, again using a 150 kilometer spatial cutoff. Directionally, the results for the focal point are consistent with the idea that compliance increases as more projects operate within the same area. The rationale is that more projects in the same area increase area legibility and make for more efficient supervision missions, enabling them to monitor multiple projects in the same trip.

The focal point results for the minimal adjustment sets are statistically significant at the 10% level but not at the 5% level. The canonical adjustment set results have even higher p-values, suggesting even further caution. Considering that compliance follows a 1-4 scale, the effect sizes are also rather small. Specifically, each additional project in the same focal point increases compliance by 0.005 points—roughly 7.8 times smaller than the log driving time estimates. The results after adding first- and second-order neighboring grid cells consistent with a Queen contiguity approach move the estimate in the opposite direction, suggesting two conclusions. First, there may be a saturation point beyond which additional nearby projects no longer help. Second, project density, which captures geography more than institutions, cannot substitute for the infrastructural power from travel times.

## 4. Robustness

To probe the robustness of the travel time results, we switch the spatial cutoff from 150 to 75 kilometers in Table C.1. The results are very similar to the main results in Table 3.

Figure 7: Project Density Results: 150 Kilometer Spatial Cutoff




We find similar results from when we adopt a 75 kilometer spatial cutoff for the interactions (Table C.1) as well as the density results (Figure D.1).

Consistent with our pre-analysis plan, we also analyze the sensitivity of the travel time results to controlling for other HYDE population measures. Specifically, we individually swap the log population counts control for log population density, log rural population

counts, log urban population counts, and log built-up area.[16] Across 16 different models in Figure C.2, the only two models that marginally lose statistical significance are the ones including the log rural population counts under the minimal adjustment set. On the one hand, rural population increases nullify remoteness and, in turn, our hypothesis about its effects. On the other hand, once we switch to the canonical adjustment set, which accounts for precision-oriented controls, the results controlling for log rural population counts regain statistical significance. These results thus suggest that the travel time results are highly robust even to the most challenging potential confounders. The results are also theoretically important given high-profile scholarly debates about the role of geography and population in international development (Herbst, 2000; Robinson, 2002; Bandyopadhyay and Green, 2012).

# 5. Conclusion

In this paper, we probe the interplay of geographic structures, institutions, and the agency of international bureaucrats in development by studying foreign aid implementation. Specifically, we examine compliance with the World Bank's safeguard policies on involuntary resettlement, indigenous peoples, and the environment, which protect people from the negative externalities of development.

Consistent with our pre-analysis plan and theory focusing on rational inattention, we find that longer driving times to reach and monitor projects decrease compliance, and effects concentrate among the most inaccessible projects. We find less support for the effects of geographic proximity on compliance, suggesting that geography cannot fully substitute for road networks and state presence—that is, the state's infrastructural power (Mann, 1984). Although structures dominate in this setting, our results suggest that international bureaucrats still have *some* agency to mitigate structural development challenges. These results are noteworthy given that travel times to reach projects represent a major structural challenge and least-likely case for international bureaucrats to impact project compliance outcomes.

[16] All of these measures correlate at 0.67 or higher (see Figure A.2), so including them all in the same regression would lead to collinearity concerns.

From a theoretical perspective, our statistical results consistent with rational inattention yield two related insights. First, they elucidate that although theories of compliance tend to draw sharp lines, integrating across them can clarify how bureaucrats behave under costly monitoring and noisy information (see also Tallberg, 2002). Second, distinctions between geography versus institutions, structure versus agency, and other phenomena that tend to provoke bifurcated arguments are often not so clear-cut. To address these challenges, the special issue framework paper offers valuable guidance: structures often abound, and the key analytic task is to distinguish whether they align, conflict, or operate orthogonally (Weaver, Morrison and Heinzel, 2026). Our case highlights conforming alignment. Geographic inaccessibility, weak state reach, and organizational incentives all push bureaucrats in the same direction, limiting international bureaucrats' agency and scope to affect outcomes.

Building on the aid targeting literature (Briggs, 2017, 2021; Marineau and Findley, 2020), our results also suggests a clear policy conclusion: there is a streetlight effect in foreign aid,[17] and it is consequential for key outcomes. Inaccessible locations are often among the places that need development programs most. Yet these locations may also require different project designs, tighter safeguards before implementation, and stronger monitoring (e.g., with help local NGOs). Managing these tensions is central to the work of foreign aid and development.

[17]See Hendrix (2017) for an overview of the streetlight effect phenomenon.

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# Appendices

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# A. Summary Statistics

Figure A.1: Correlation Plot

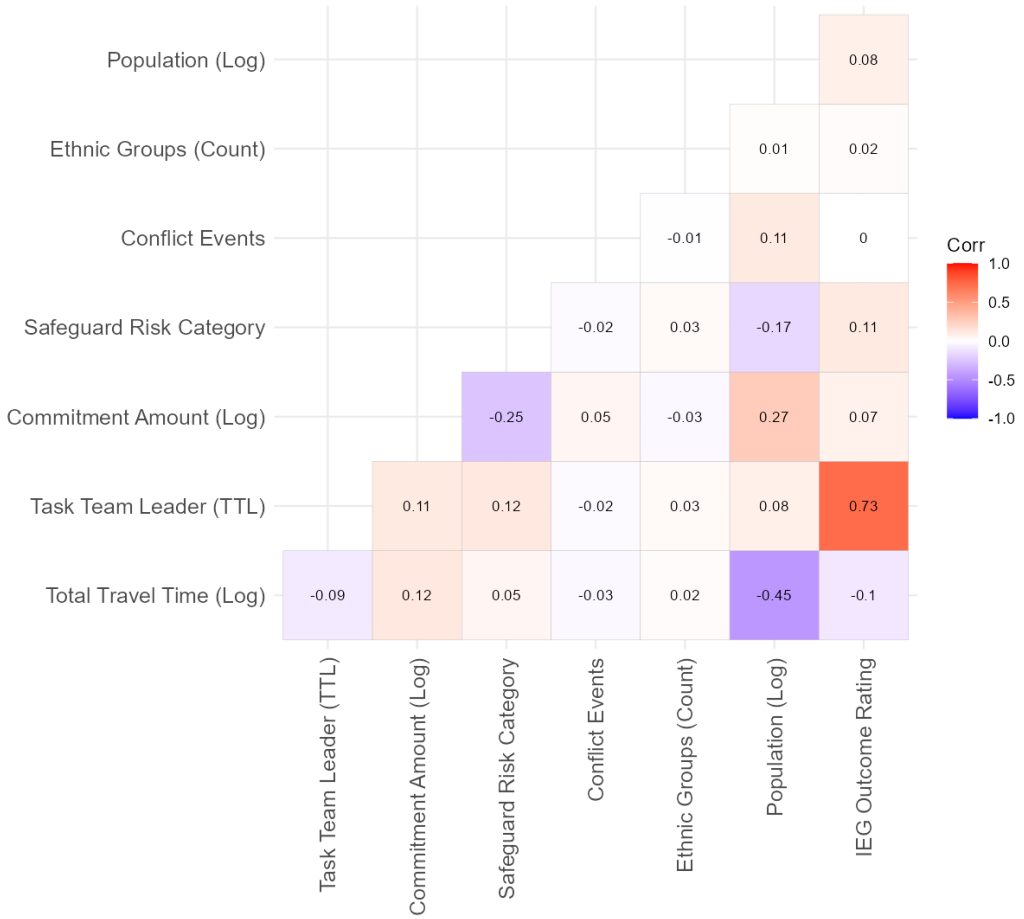



Table A.1: 150 Balance Table: Full vs TTL Sample

| Variable | Mean.NonTTL | Mean.TTL | Std.Diff | Var.Ratio | P.Value |
|---|---|---|---|---|---|
| Commitment (Log) | 18.315 | 18.607 | 0.261 | 0.887 | <0.001 |
| Safeguard Category | 1.948 | 1.806 | -0.295 | 1.201 | <0.001 |
| Conflict Count | 0.182 | 0.111 | -0.133 | 0.527 | <0.001 |
| Ethnic Groups | 0.498 | 0.523 | 0.032 | 1.099 | <0.001 |
| Population (Log) | 8.829 | 8.969 | 0.074 | 0.964 | <0.001 |

Figure A.2: Correlation Plot (with Extra Population Measures)

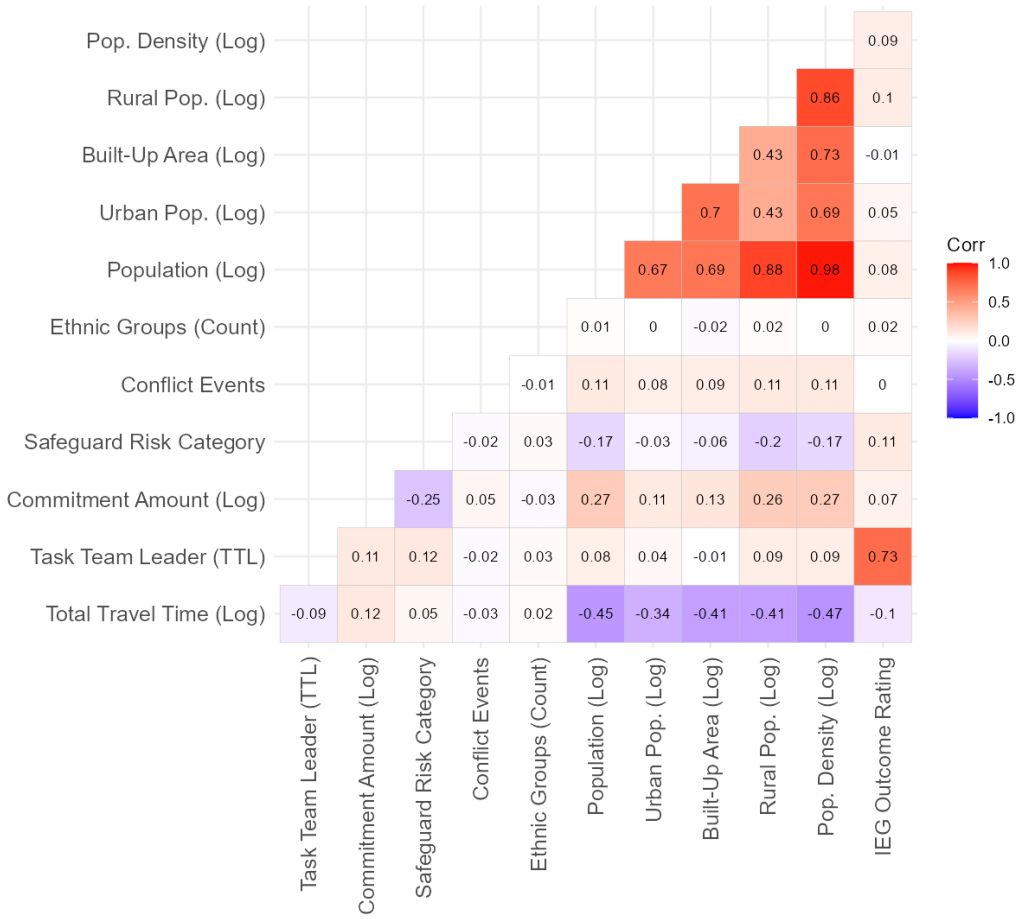



Table A.2: 75 Balance Table: Full vs TTL Sample

| Variable | Mean.NonTTL | Mean.TTL | Std.Diff | Var.Ratio | P.Value |
|---|---|---|---|---|---|
| Commitment (Log) | 18.315 | 18.607 | 0.261 | 0.887 | <0.001 |
| Safeguard Category | 1.948 | 1.806 | -0.295 | 1.201 | <0.001 |
| Conflict Count | 0.182 | 0.111 | -0.133 | 0.527 | <0.001 |
| Ethnic Groups | 0.498 | 0.523 | 0.032 | 1.099 | <0.001 |
| Population (Log) | 8.829 | 8.969 | 0.074 | 0.964 | <0.001 |

Table A.3: Balance Table Describing the Full Sample vs. TTL-Specific Sample

| Variable | Full Mean | Full SD | TTL Mean | TTL SD | *p*-value | SMD |
|---|---|---|---|---|---|---|
| TTL | 4.082 | 0.592 | 4.201 | 0.538 | <0.001 | -0.210 |
| Safeguard Category | 2.056 | 0.723 | 1.924 | 0.846 | <0.001 | 0.168 |
| Commitment (Log) | 18.323 | 1.146 | 18.659 | 1.088 | <0.001 | -0.300 |
| Ethnic Groups | 2.110 | 1.335 | 2.181 | 1.377 | <0.001 | -0.053 |
| Conflict Count | 0.149 | 0.532 | 0.103 | 0.418 | <0.001 | 0.096 |
| Population (Log) | 8.871 | 1.930 | 9.367 | 1.769 | <0.001 | -0.268 |

Table A.4: Summary Statistics for Table 3, Column (3)

| Variable | mean | sd | min | p1 | p5 | p10 | p25 | p50 | p75 | p90 | p95 | p99 | max |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Compliance (Spatial) | 3.23 | 0.85 | 1.00 | 1.00 | 2.00 | 2.00 | 3.00 | 3.00 | 4.00 | 4.00 | 4.00 | 4.00 | 4.00 |
| Travel Time (Log) | 5.87 | 0.99 | 0.00 | 3.08 | 4.14 | 4.69 | 5.30 | 5.92 | 6.49 | 7.08 | 7.34 | 8.06 | 10.48 |
| Travel Time (Minutes) | 564.98 | 1038.76 | 0.00 | 20.70 | 62.10 | 107.60 | 199.95 | 372.10 | 656.70 | 1191.80 | 1541.50 | 3178.60 | 35 715.40 |
| TTL × Travel Time (Log) | 24.03 | 5.10 | 0.00 | 11.02 | 15.46 | 17.71 | 20.87 | 24.06 | 27.38 | 30.50 | 32.36 | 35.22 | 51.89 |
| TTL | 4.10 | 0.58 | 1.69 | 2.45 | 3.10 | 3.35 | 3.77 | 4.10 | 4.52 | 4.82 | 4.95 | 5.21 | 5.97 |
| Commitment (Log) | 18.38 | 1.13 | 13.74 | 15.55 | 16.59 | 16.91 | 17.61 | 18.45 | 19.21 | 19.72 | 20.06 | 21.21 | 22.13 |
| Safeguard Category | 1.92 | 0.48 | 1.00 | 1.00 | 1.00 | 1.00 | 2.00 | 2.00 | 2.00 | 2.00 | 3.00 | 3.00 | 3.00 |
| Conflict Count | 0.13 | 0.49 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 | 2.00 | 8.00 |
| Ethnic Groups | 0.54 | 0.83 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 | 2.00 | 2.00 | 3.00 | 8.00 |
| Population (Log) | 9.03 | 1.86 | 0.00 | 3.42 | 5.55 | 6.74 | 8.04 | 9.23 | 10.31 | 11.19 | 11.54 | 12.54 | 13.58 |

# B. DAG Paths

In this section, we describe the paths underpinning all of the coding decisions in the DAG (see Figure 5):

- **Travel time/density**:
  - *Civil conflict* affects travel time (or project density) because conflicts make travel more difficult. Beyond increasing security risks, civil conflicts can result in the destruction of road infrastructure that makes travel slower.
  - *Population* affects travel time (or project density) because cities generally have greater access to airports, which facilitate travel for bureaucrats conducting supervision missions from abroad.
- **Population**:
  - *Civil conflict* affects population because citizens wish to reside in areas without conflict for security reasons.
- **Safeguard category**:
  - *Population* affects the safeguard risk category because population affects the ability of projects to receive goods, services, and other measures that may reduce risks World Bank (2013, 2022).
  - *Civil conflict* affects safeguard risk category because civil conflict makes assistance and other services more difficult to furnish in the case of a safeguard dispute.
- **TTL performance**:
  - *Commitments* affect TTL performance because the World Bank generally will not assign an inexperienced or lower-performing TTL to projects involving larger commitment amounts. The relationship flows from commitments to TTL, not the other way around, because the World Bank determines commitment amounts according to country-level criteria (Denly, 2021).

# C. Additional Travel Time Results

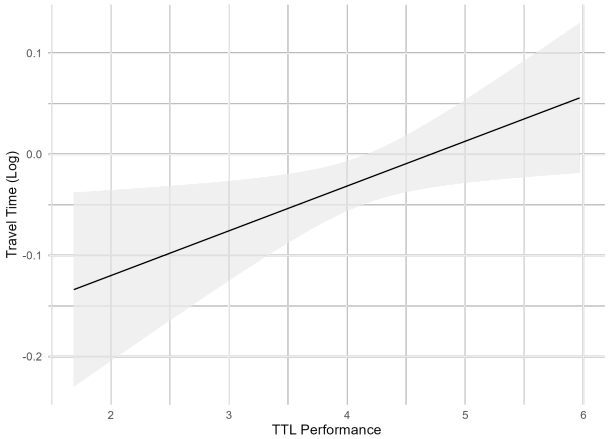
Table C.1: Travel Time and the Interactive Effect of TTL Performance on Compliance (75 KM Spatial Cutoff)

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| Total Travel Time (Log) | −0.032** | −0.032** | −0.039*** | −0.039*** | −0.208** | −0.208** | −0.195** | −0.196** |
| | (0.014) | (0.014) | (0.014) | (0.014) | (0.081) | (0.081) | (0.089) | (0.089) |
| Travel Time × TTL (Log) | | | | | 0.044** | 0.044** | 0.038* | 0.039* |
| | | | | | (0.019) | (0.019) | (0.021) | (0.021) |
| TTL | | | | | 0.015 | 0.015 | 0.018 | 0.017 |
| | | | | | (0.117) | (0.117) | (0.128) | (0.128) |
| Commitment (Log) | | | −0.063*** | −0.066*** | −0.158*** | −0.158*** | −0.119*** | −0.117*** |
| | | | (0.019) | (0.019) | (0.021) | (0.021) | (0.021) | (0.020) |
| Safeguard Category | | | 0.245*** | 0.244*** | | | 0.236*** | 0.236*** |
| | | | (0.034) | (0.034) | | | (0.038) | (0.038) |
| Conflict Count | 0.023 | 0.022 | 0.016 | 0.015 | | | 0.000 | 0.003 |
| | (0.019) | (0.019) | (0.019) | (0.019) | | | (0.021) | (0.021) |
| Ethnic Groups | | | 0.008 | 0.008 | | | 0.010 | 0.012 |
| | | | (0.015) | (0.014) | | | (0.016) | (0.016) |
| Population (Log) | −0.009 | −0.009 | −0.009 | −0.009 | | | −0.010 | −0.009 |
| | (0.009) | (0.009) | (0.009) | (0.009) | | | (0.010) | (0.010) |
| Observations | 101 264 | 101 264 | 97 308 | 97 308 | 89 276 | 89 276 | 85 180 | 85 180 |
| $R^2$ | 0.190 | 0.191 | 0.212 | 0.214 | 0.241 | 0.243 | 0.254 | 0.255 |
| Adj. $R^2$ | 0.189 | 0.190 | 0.211 | 0.212 | 0.240 | 0.241 | 0.252 | 0.254 |
| Country FE | X | | X | | X | | X | |
| Country + Year FE | | X | | X | | X | | X |

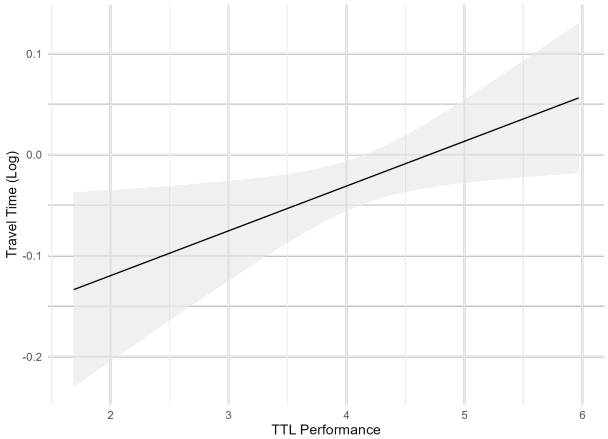
Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; spatial cutoff at 75 kilometers; minimal adjustment sets in (1), (2), (5), and (6); canonical adjustment sets in (3), (4), (7), and (8).

Figure C.1: Marginal Effects for Interactions (75 Kilometer Spatial Cutoff)

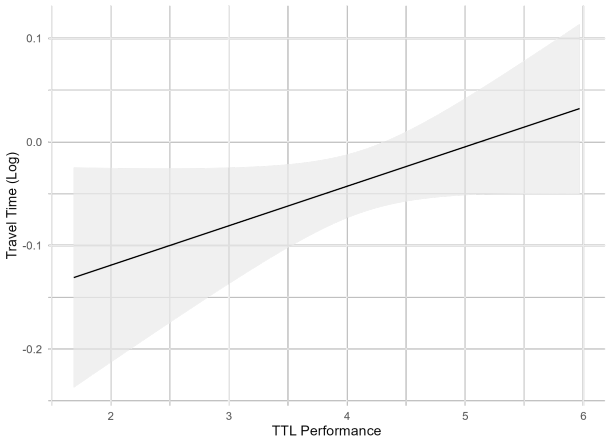
(a) Minimal Adjustment Set (Country FE)



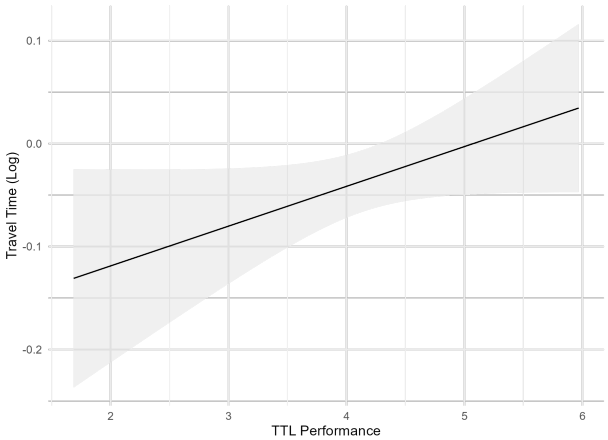

(b) Minimal Adjustment Set (Country + Year FE)




(c) Canonical Adjustment Set (Country FE)

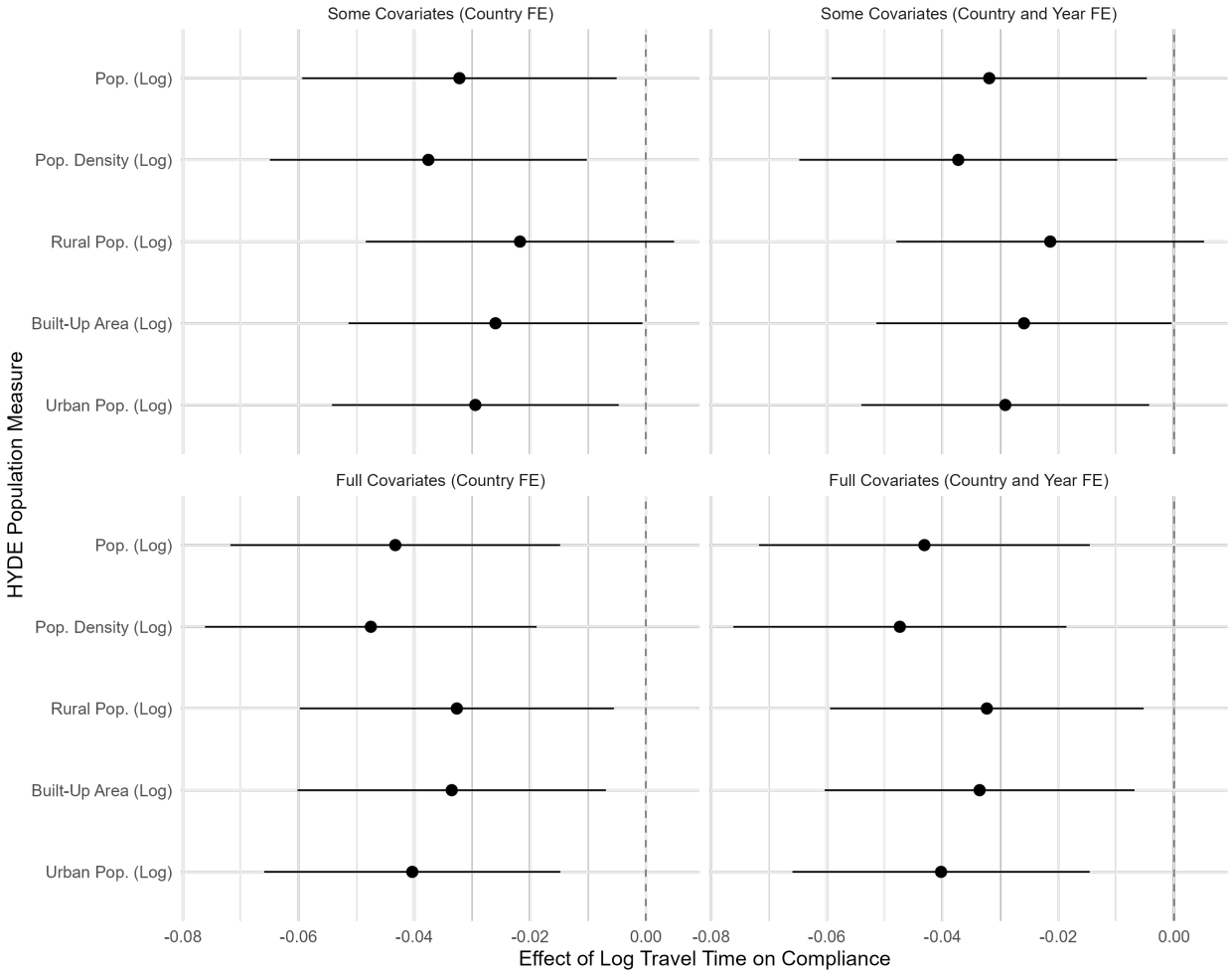



(d) Canonical Adjustment Set (Country + Year FE)




Note: the above estimates reflect 95% confidence intervals.

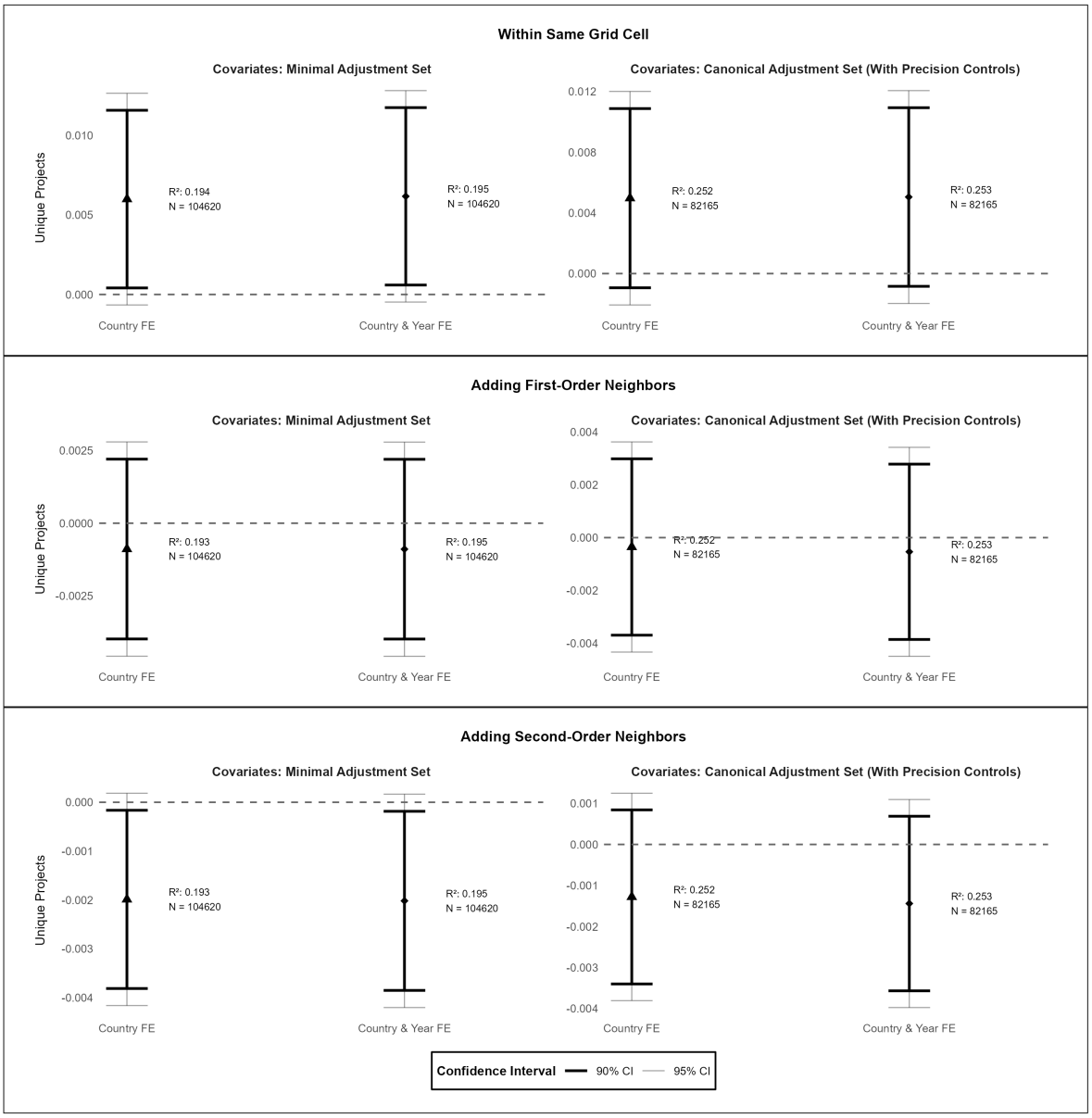
Figure C.2: Robustness to Controlling for Different Population Measures




Note: 95% confidence intervals for spatial HAC models with 150 kilometer spatial cutoffs.

# D. Additional Density Results

Figure D.1: Density Plot: 75 Kilometer Cutoff



# E. TTL-Specific Analyses

As we specified in Section 2.4.2, only 17% of the project-cell-year observations include TTLs supervising multiple projects within the same country. These observations are also not a random sample of the full sample (see Table A.3), suggesting series selection-related problems that are likely worthy of a whole new paper. We only report the results here to be consistent with our pre-analysis plan.

Table E.1: TTL-Specific Travel Time and the Interactive Effect of TTL Performance on Compliance (150 KM Spatial Cutoff)

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| Total Travel Time (Log) | −0.018 | −0.019 | −0.032 | −0.032 | −0.002 | −0.004 | 0.025 | 0.025 |
| | (0.022) | (0.022) | (0.024) | (0.025) | (0.141) | (0.141) | (0.163) | (0.162) |
| TTL × Travel Time (Log) | | | | | −0.009 | −0.008 | −0.013 | −0.013 |
| | | | | | (0.034) | (0.034) | (0.039) | (0.039) |
| TTL | | | | | 0.328 | 0.324 | 0.317 | 0.316 |
| | | | | | (0.228) | (0.227) | (0.263) | (0.261) |
| Commitment (Log) | | | −0.114*** | −0.114*** | −0.160*** | −0.161*** | −0.125*** | −0.123*** |
| | | | (0.036) | (0.036) | (0.033) | (0.033) | (0.036) | (0.036) |
| Safeguard Category | | | 0.213*** | 0.214*** | | | 0.168*** | 0.170*** |
| | | | (0.060) | (0.059) | | | (0.062) | (0.062) |
| Conflict Count | 0.048 | 0.043 | 0.038 | 0.033 | | | 0.039 | 0.035 |
| | (0.049) | (0.049) | (0.044) | (0.044) | | | (0.042) | (0.042) |
| Ethnic Groups | | | −0.006 | −0.007 | | | −0.009 | −0.008 |
| | | | (0.029) | (0.029) | | | (0.029) | (0.029) |
| Population (Log) | 0.020 | 0.020 | 0.013 | 0.013 | | | 0.014 | 0.015 |
| | (0.013) | (0.013) | (0.013) | (0.013) | | | (0.014) | (0.014) |
| Observations | 27 018 | 27 018 | 25 855 | 25 855 | 27 150 | 27 150 | 25 855 | 25 855 |
| $R^2$ | 0.303 | 0.305 | 0.330 | 0.332 | 0.334 | 0.336 | 0.343 | 0.345 |
| $Adj.R^2$ | 0.300 | 0.302 | 0.327 | 0.329 | 0.332 | 0.334 | 0.340 | 0.342 |
| Country FE | X | | X | | X | | X | |
| Country + Year FE | | X | | X | | X | | X |

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; spatial cutoff at 150 kilometers; minimal adjustment sets in (1), (2), (5), and (6); canonical adjustment sets in (3), (4), (7), and (8).

Table E.2: TTL-Specific Travel Time and the Interactive Effect of TTL Performance on Compliance (75 KM Spatial Cutoff)

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| Total Travel Time (Log) | −0.018 | −0.019 | −0.032 | −0.032 | −0.002 | −0.004 | 0.025 | 0.025 |
| | (0.020) | (0.020) | (0.023) | (0.023) | (0.131) | (0.131) | (0.147) | (0.147) |
| TTL × Travel Time (Log) | | | | | −0.009 | −0.008 | −0.013 | −0.013 |
| | | | | | (0.031) | (0.031) | (0.035) | (0.035) |
| TTL | | | | | 0.328 | 0.324 | 0.317 | 0.316 |
| | | | | | (0.206) | (0.206) | (0.233) | (0.232) |
| Commitment (Log) | | | −0.114*** | −0.114*** | −0.160*** | −0.161*** | −0.125*** | −0.123*** |
| | | | (0.032) | (0.032) | (0.030) | (0.030) | (0.032) | (0.032) |
| Safeguard Category | | | 0.213*** | 0.214*** | | | 0.168*** | 0.170*** |
| | | | (0.052) | (0.052) | | | (0.055) | (0.055) |
| Conflict Count | 0.048 | 0.043 | 0.038 | 0.033 | | | 0.039 | 0.035 |
| | (0.046) | (0.046) | (0.042) | (0.041) | | | (0.040) | (0.040) |
| Ethnic Groups | | | −0.006 | −0.007 | | | −0.009 | −0.008 |
| | | | (0.025) | (0.025) | | | (0.025) | (0.025) |
| Population (Log) | 0.020 | 0.020 | 0.013 | 0.013 | | | 0.014 | 0.015 |
| | (0.012) | (0.012) | (0.013) | (0.013) | | | (0.013) | (0.013) |
| Observations | 27 018 | 27 018 | 25 855 | 25 855 | 27 150 | 27 150 | 25 855 | 25 855 |
| $R^2$ | 0.303 | 0.305 | 0.330 | 0.332 | 0.334 | 0.336 | 0.343 | 0.345 |
| $Adj.R^2$ | 0.300 | 0.302 | 0.327 | 0.329 | 0.332 | 0.334 | 0.340 | 0.342 |
| Country FE | X | | X | | X | | X | |
| Country + Year FE | | X | | X | | X | | X |

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; spatial cutoff at 75 kilometers; minimal adjustment sets in (1), (2), (5), and (6); canonical adjustment sets in (3), (4), (7), and (8).

Figure E.1: Density Plot: 150 km Cutoff (TTL-Specific)

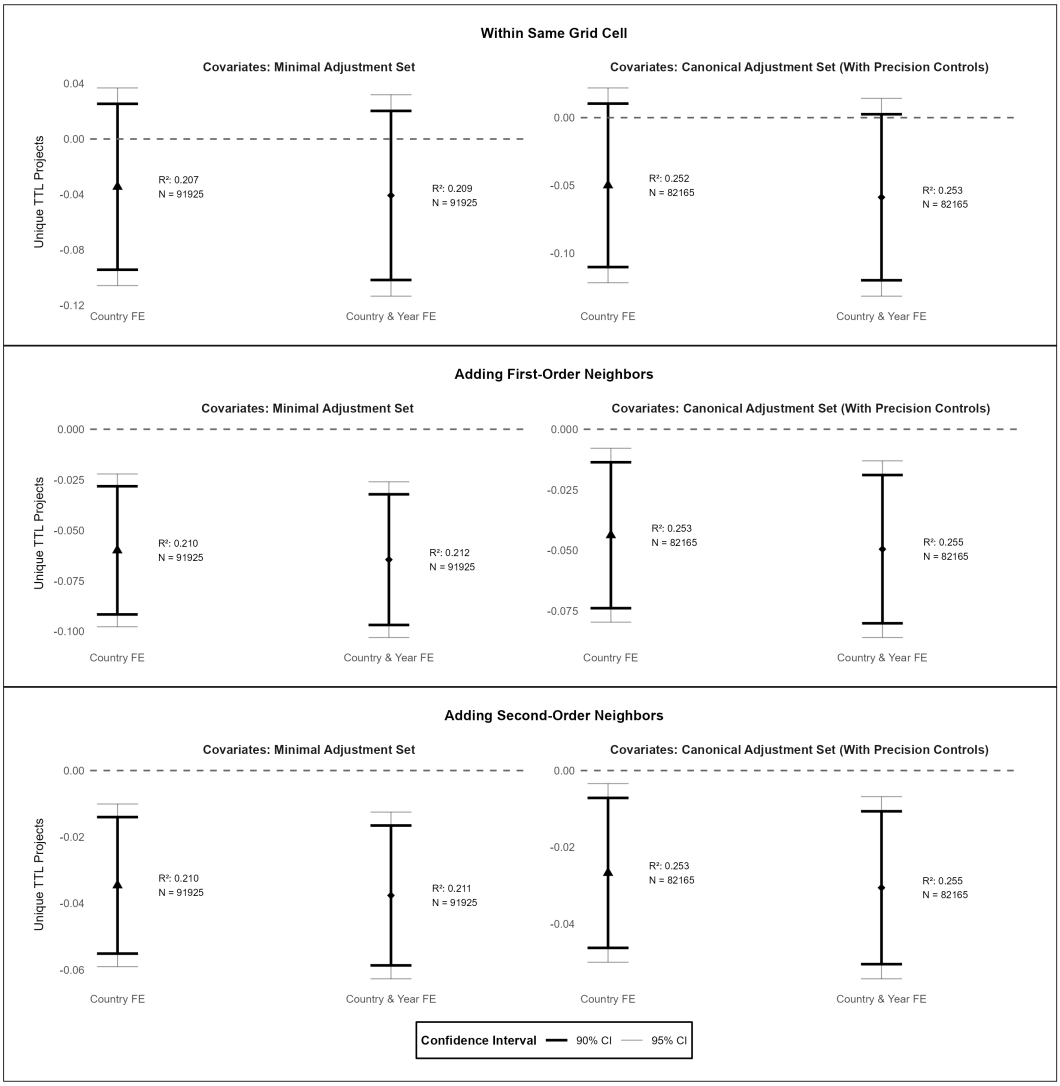
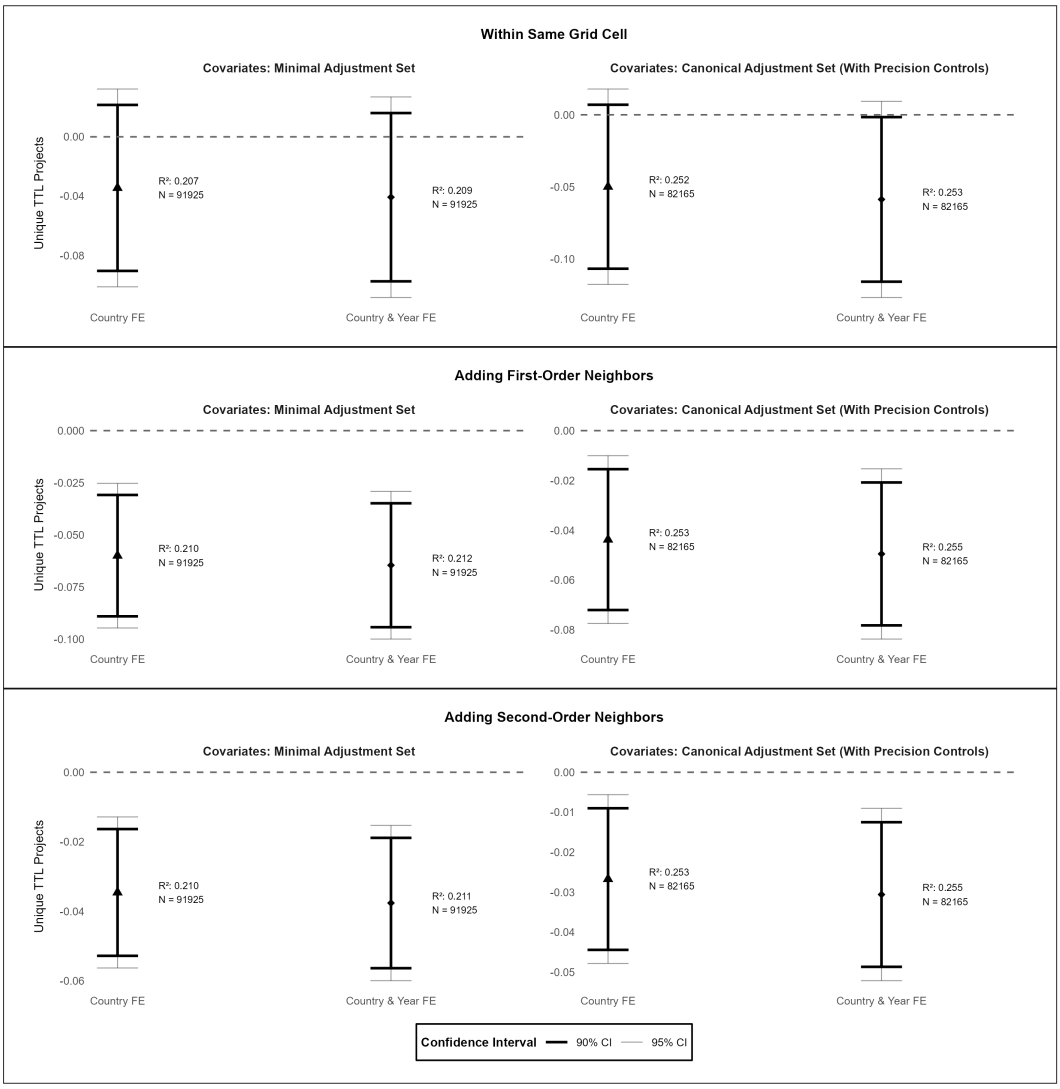


Figure E.2: Density Plot: 75 km Cutoff (TTL-Specific)



# F. Mathematical Framework

## F.1. Input Data

Let:

- $P = \{p_1, p_2, \ldots, p_n\}$ be the set of project locations for a given project, where each $p_i$ refers to the latitude and longitude coordinates of each project's location.
- $A = \{a_1, a_2, \ldots, a_m\}$ be the set of airports in the same country, where each $a_j$ uses data from Megginson (2025) to denote each airport's latitude and longitude coordinates as well as its size in terms of whether it is `large`, `medium`, or `small`.
- $O = \{o_1, o_2, \ldots, o_k\}$ be the set of other project locations in the same country and year, excluding the current project, where $o_i$ refers to the latitude and longitude coordinates of each other project's location.

## F.2. Nearest Airport Selection

For each project, the algorithm selects one airport $a^* \in A$ that minimizes the total travel time to all project locations $P$ while ensuring at least one non-zero travel time. The algorithm evaluates airports in the following priority order: `large`, then `medium`, and finally `small`.

We define the travel time matrix $D$ from OSRM as:

$$D_{j,i} = \text{travel time (seconds) from } a_j \text{ to } p_i$$

We compute the total travel time for airport $a_j$ by

$$S_j = \sum_{i=1}^{n} D_{j,i}$$

Then, we select the airport $a^*$ as:

$$a^* = \arg\min_{a_j \in A_{\text{size}}} S_j \quad \text{such that} \quad S_j > 0$$

where $A_{\text{size}}$ represents the subset of airports for the current size being evaluated. The minimization first considers `large` airports, then `medium`, and finally `small` until a valid $a^*$ is found. If no $a^*$ with $S_j > 0$ exists, the related fields are set to missing. For each location $p_i$, if $a^*$ is selected, we compute:

$$\text{travel time in minutes}_i = \frac{D_{*,i}}{60}$$

where $*$ denotes the index of $a^*$, respectively. If $D_{*,i} = 0$ or if no $a^*$ is found, we assign it as missing.

### F.3. Neighbor Calculations

For each location $p_i$, we identify the nearest neighbor project $o^* \in O$ within the same country. We define the neighbor travel time $N$ from OSRM as:

$$N_{i,l} = \text{travel time (seconds) from } p_i \text{ to } o_l$$

We next determine the nearest general neighbor, $o^*$, by

$$o^* = \arg\min_{o_l \in O} N_{i,l}$$

Then, we calculate the neighbor time as follows:

$$\text{neighbor time} = \frac{N_{i,*}}{60}$$

Next, we compute the total travel time in minutes as follows:

$$\text{total time minutes}_i = \text{travel time in minutes}_i + \text{neighbor time}$$

## G. Local Server Setup and Data Processing Steps

### G.1. Overview

The Open Source Routing Machine (OSRM) is a high-performance routing engine that computes shortest or fastest paths across large-scale road networks using OpenStreetMap (OSM) data. It employs the Multi-Level Dijkstra (MLD) algorithm to enable efficient routing queries, making it suitable for applications requiring rapid pathfinding on continental-scale datasets. This appendix provides a comprehensive, step-by-step guide to installing prerequisites, preparing OSM data, building routing graphs, and launching a local OSRM server on a Windows system using Windows Subsystem for Linux 2 (WSL2) and Docker.

We assume that regional `.osm.pbf` files, which are Protocol Buffer Format files containing OSM data, have been downloaded from Geofabrik (OpenStreetMap Foundation, 2025), a service providing pre-processed OSM extracts by continent or country, and are stored locally. This guide is tailored for Windows users and prioritizes reproducibility: all variables are explicitly declared, commands are formatted for direct copy-pasting into PowerShell, and detailed explanations clarify each step's purpose and potential challenges. Troubleshooting tips and resource considerations are included to address common issues, particularly for memory-intensive tasks like preprocessing large datasets.

### G.2. System Requirements

Processing large OSM files (e.g., continent-scale datasets exceeding 10 GB) requires substantial computational resources. The following are recommended:

- Windows 10 (version 2004 or later) or Windows 11 with administrator privileges to enable system features like WSL2.
- A solid-state drive (SSD) with at least 500 GB of free space, as preprocessing generates temporary files that can be 5–10 times the size of the input `.osm.pbf` file.
- At least 32 GB of RAM (64 GB or more preferred) to handle memory-intensive graph partitioning and customization steps.
- A multi-core CPU (at least 8 cores, 12 or more preferred) for parallel processing during graph construction.
- WSL2 enabled for Linux compatibility and Docker Desktop installed for containerized execution of OSRM tools.

If your system does not meet these specifications, consider using smaller regional datasets (e.g., individual countries) or cloud-based alternatives like AWS EC2 instances with sufficient resources.

## G.3. Step 1: Enable WSL2

Windows Subsystem for Linux 2 (WSL2) provides a lightweight Linux kernel within Windows, offering superior performance and compatibility for tools like Docker compared to WSL1. To enable WSL2:

Open **PowerShell as Administrator** (right-click the PowerShell icon and select "Run as administrator"):

```
wsl --install
```

This command installs WSL2 and the default Linux distribution (Ubuntu recommended for its stability and package ecosystem). After execution, Windows will prompt for a system reboot. Following the reboot, complete the Ubuntu setup by setting a username and password when prompted.

Verify that WSL2 is correctly installed:

```
wsl --version
# Expected output: "WSL version: 2" along with kernel and distribution details.
```

If the output indicates WSL1, convert the Ubuntu distribution to WSL2:

```
wsl --set-version Ubuntu 2
```

**Troubleshooting**: If `wsl -install` fails, ensure virtualization is enabled in your system's BIOS/UEFI settings (look for "Virtualization," "VT-x," or "AMD-V" in your motherboard manual). In Windows, verify that the "Virtual Machine Platform" and "Windows

Subsystem for Linux" features are enabled via *Control Panel* → *Programs* → *Turn Windows features on or off*. If issues persist, update Windows to the latest version.

## G.4. Step 2: Install Docker Desktop

Docker Desktop provides a user-friendly interface for managing containers on Windows and integrates seamlessly with WSL2 to run Linux-based images like OSRM.

1. Download the Docker Desktop installer from https://www.docker.com/products/docker-desktop and run it with administrator privileges. Follow the on-screen instructions to complete the installation.
2. After installation, open Docker Desktop and configure the following settings for optimal performance:
   - In *Settings* → *General*, ensure **Use the WSL 2 based engine** is checked. This leverages WSL2 for better resource isolation and performance.
   - Under *Resources* → *WSL Integration*, enable integration with your Ubuntu distribution to allow Docker commands to run within WSL2.
   - Under *Resources* → *Advanced*, allocate at least 32 GB of RAM and 12 CPU threads to Docker. This prevents out-of-memory errors during OSRM preprocessing. Adjust based on your hardware, but leave sufficient resources for Windows to avoid system instability.
   - Under *Resources* → *File Sharing*, add your working directory (e.g., `D:\OSRM`) to allow Docker to mount it as a volume for data access.
3. Restart Docker Desktop after applying changes to ensure they take effect.

Verify Docker installation by running a test container:

```
docker run hello-world
# Expected output: A welcome message indicating Docker is functioning correctly.
```

**Troubleshooting**: If the `hello-world` container fails to run, ensure Docker Desktop is running and WSL2 integration is enabled. Check that your WSL2 distribution is listed under *WSL Integration*. If errors persist, restart Docker Desktop or reinstall it, ensuring all prerequisites (e.g., WSL2) are met.

## G.5. Step 3: Configure Swap Space for WSL2

Preprocessing large `.osm.pbf` files (e.g., continent-scale datasets) can exceed physical RAM, leading to crashes. Configuring a swap file provides virtual memory to prevent out-of-memory errors.

Create a dedicated directory for the swap file:

```
mkdir C:\wsl_swp
```

Edit the WSL2 configuration file to allocate swap space. Open `%UserProfile%\.wslconfig` (typically `C:\Users\YourUsername\.wslconfig`) in a text editor (e.g., Notepad). If the file does not exist, create it. Add the following configuration:

```
[wsl2]
memory=32GB
processors=12
swap=200GB
swapFile=C:\\wsl_swp\\wsl.swap
localhostForwarding=true
```

This configuration allocates 32 GB of RAM, 12 CPU threads, and a 200 GB swap file. Adjust `memory` and `processors` based on your hardware, and set `swap` to at least 2–3 times the size of the input `.osm.pbf` file for large datasets.

Apply the changes by shutting down WSL2:

```
wsl --shutdown
```

Restart Docker Desktop to ensure the new configuration is applied. Verify the swap file by running a Linux command within WSL2:

```
wsl free -h
# Expected output: Displays total, used, and free swap space (e.g., 200G total).
```

**Troubleshooting**: If the swap file is not recognized, ensure the `swapFile` path uses double backslashes (`\\`) and that the directory `C:\wsl_swp` exists. Avoid placing the swap file on a mechanical hard drive, as it significantly slows down preprocessing.

## G.6. Step 4: Prepare Working Directory and Data

Choose a persistent working directory on an SSD to store OSM data and processed files. SSDs are critical for performance due to the high I/O demands of preprocessing.

Set up the directory in PowerShell:

```
$OSRM_DATA = "D:\OSRM"
mkdir $OSRM_DATA
```

Download `.osm.pbf` files from Geofabrik (https://download.geofabrik.de/) and place them in the working directory. For example:

```
D:\OSRM\africa-latest.osm.pbf
D:\OSRM\asia-latest.osm.pbf
```

Ensure the directory path has no spaces or special characters, as these can cause issues with Docker volume mounting. If you encounter permission errors, verify that your user account has full read/write access to the directory.

## G.7. Step 5: Preprocess Map Data

Preprocessing transforms the raw `.osm.pbf` file into a routing graph optimized for the MLD algorithm. This involves three stages: extraction, partitioning, and customization. Each stage is executed within the OSRM Docker container to ensure a consistent environment.

Define variables for the dataset to process:

```
$OSRM_DATA = "D:\OSRM"
$PbfFile   = "africa-latest.osm.pbf"
$BaseName  = "africa-latest"
```

These variables specify the working directory, input file, and base name for output files. Ensure `$PbfFile` matches the exact filename of your downloaded `.osm.pbf` file.

Run the preprocessing stages:

**1) Extract road network with profile:** The extraction step filters the `.osm.pbf` file to extract road network data using a Lua profile (`car.lua`) that defines routing rules (e.g., speed limits, road types).

```
docker run --rm -t -v "|$OSRM_DATA|:/data" osrm/osrm-backend `
  osrm-extract -p /opt/car.lua "/data/|$PbfFile|"
```

The `-rm` flag ensures the container is removed after execution to save disk space, `-t` allocates a terminal for output, and `-v "${OSRM_DATA}:/data"` mounts the working directory to `/data` in the container. This step generates an `.osrm` file (e.g., `africa-latest.osrm`).

**2) Partition graph for MLD algorithm:** The partitioning step divides the road network into hierarchical levels to optimize query performance for the MLD algorithm.

```
docker run --rm -t -v "${OSRM_DATA}:/data" osrm/osrm-backend `
  osrm-partition "/data/$BaseName.osrm"
```

This step creates additional files, such as `.osrm.partition` and `.osrm.cells`, which store the hierarchical graph structure.

**3) Customize weights (speed limits, turn costs):** The customization step applies weights (e.g., travel times, turn penalties) to the partitioned graph, enabling fast routing queries.

```
docker run --rm -t -v "${OSRM_DATA}:/data" osrm/osrm-backend `
  osrm-customize "/data/$BaseName.osrm"
```

This step generates files like `.osrm.mldgr`, completing the preprocessing pipeline.

Upon completion, the `D:\OSRM` directory will contain derived files, including:

- `africa-latest.osrm`
- `africa-latest.osrm.nodes`
- `africa-latest.osrm.edges`
- `africa-latest.osrm.mldgr`
- `africa-latest.osrm.partition`
- `africa-latest.osrm.cells`

**Troubleshooting**: Preprocessing can fail due to insufficient memory or disk space. Monitor resource usage with tools like Task Manager or `htop` in WSL2. If errors occur, verify that the swap file is active and the working directory is accessible. For very large datasets, consider increasing the swap size or using a smaller region. If Docker commands fail, ensure Docker Desktop is running and the volume mount path is correct.

## G.8. Step 6: Launch Routing Server

Start the OSRM server to handle routing queries using the preprocessed data. First, remove any existing container to avoid conflicts:

```
docker rm -f osrm 2>$null
```

The `-f` flag forces removal, and `2>$null` suppresses error messages if no container exists. Then, launch the server:

```
docker run -d --name osrm -p 5000:5000 -v "${OSRM_DATA}:/data" osrm/osrm-backend `
  osrm-routed --algorithm mld "/data/$BaseName.osrm"
```

The `-d` flag runs the container in detached mode (background), `-name osrm` names the container, `-p 5000:5000` maps port 5000 on the host to 5000 in the container, and `-algorithm mld` specifies the MLD algorithm for routing.

## G.9. Step 7: Verify Server Health

Confirm that the server is running correctly by checking its health endpoint:

```
Invoke-RestMethod "http://localhost:5000/health"
# Expected response: OK
```

Test the server with a sample routing query, such as finding the shortest driving path between two coordinates (e.g., Houston, TX at -95.3698, 29.7604 and Dallas, TX at -96.7969, 32.7767):

```
Invoke-RestMethod "http://localhost:5000/route/v1/driving/-95.3698,29.7604;-96.7969,32.7767
    "
```

This returns a JSON object containing the lat and longitude of said project,airport,and nearest location ass well as the distance, and estimated travel time. For example, the response includes a `polyline` field encoding the route path and a `duration` field in seconds.

This workflow converts OSRM's returned `duration` (seconds) to `travel_time_minutes` (minutes) by dividing by 60, and converts `distance` (meters) to `travel_distance_km` (kilometers) by dividing by 1000; these converted values are what get written to the final CSV.

**Troubleshooting**: If the health endpoint returns an error or no response, verify that the container is running (`docker ps`) and that port 5000 is not blocked by a firewall. If the routing query fails, ensure the coordinates are within the preprocessed region's bounds and that preprocessing completed successfully.

## G.10. Step 8: Switching Regions

To switch to a different preprocessed region (e.g., from Africa to Asia), use a PowerShell script to automate stopping the current OSRM server, removing its container, and launching a new server with the desired dataset. The script `Switch-OSRMRegion.ps1` simplifies this process by taking the dataset's base name (e.g., `asia-latest`) as a parameter.

**Prepare the Script:** Save the following PowerShell script as `Switch-OSRMRegion.ps1` in your working directory (e.g., `D:\OSRM`):

```
# Switch-OSRMRegion.ps1
# Switches the OSRM server to a new preprocessed region dataset

param (
    [Parameter(Mandatory=$true)]
    [string]$BaseName
)

# Define working directory
$OSRM_DATA = "D:\OSRM"
```

```
# Stop and remove existing OSRM container
docker stop osrm 2>|$\|null
docker rm osrm 2>|$\|null

# Launch new OSRM server with specified dataset
docker run -d --name osrm -p 5000:5000 -v "|$OSRM_DATA|:/data" osrm/osrm-backend `
    osrm-routed --algorithm mld "/data/|$BaseName|.osrm"

Write-Output "OSRM server switched to $BaseName dataset."
```

Ensure the dataset for the new region (e.g., `asia-latest.osrm`) has been preprocessed as described in Section G (Step 5).

**Run the Script:** Execute the script in PowerShell, specifying the base name of the preprocessed dataset:

```
Put script name here
```

This command stops any running OSRM container, removes it, and starts a new server using the `asia-latest.osrm` dataset. The script assumes the working directory is `D:\OSRM`; adjust the `${OSRM_DATA}` variable in the script if your directory differs.

**Troubleshooting**: If the script fails, verify that Docker Desktop is running and the specified `.osrm` file exists in the working directory. Ensure the dataset has been fully preprocessed (including extraction, partitioning, and customization). If the port 5000 is in use, check for conflicting processes with `netstat -a -n -o | find "5000"` and terminate them if necessary.

Ensure the new `$BaseName` matches the preprocessed dataset's base filename. Preprocessing must be completed for the new region beforehand, following the steps in Section G (Step 5).

## G.11. Summary

This appendix provides a fully reproducible pipeline for setting up and running a local OSRM routing server on Windows using WSL2 and Docker. By explicitly declaring variables and file paths, and including detailed instructions and troubleshooting tips, this guide enables readers to replicate the computational environment with their own Geofabrik datasets. The pipeline supports large-scale routing applications and can be adapted to different regions or extended with custom Lua profiles for specialized routing scenarios.